



A Holistic Review of Agentic AI Frameworks, Applications, and Research Trajectories

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Abstract

Agentic Artificial Intelligence (Agentic AI), as a new generation of intelligent systems, extends beyond mere text or image generation by incorporating components such as multi step reasoning, persistent memory, multi agent interaction, and purposeful tool use. These features enable autonomy and dynamic decision making in open and complex environments. This paper provides an extensive review of the existing literature on Agentic AI. To this end, the conceptual and historical distinctions between Agentic Artificial Intelligence, classical agents, and general-purpose language models are first examined. Subsequently, the proposed architectural framework encompassing perception, role adoption, memory, planning and reflection, action and tool use, and online learning is elaborated. The discussion then turns to key infrastructures, including state and memory management, orchestration and Machine Learning Operations (MLOps), as well as agent safety. Evaluation metrics and protocols are reviewed, ranging from interactive and long horizon testing to process level assessments. Furthermore, applications of Agentic AI are introduced in areas such as scientific discovery, intelligent education, finance, and robotics, followed by an analysis of core challenges including emergent behaviors, multi agent security, accountability, and energy sustainability. The findings of this review suggest that integrating methods such as Agentic Retrieval Augmented Generation (Agentic RAG), agent chaining, and formal threat modeling can provide the foundation for the next generation of trustworthy and scalable systems. Finally, research gaps and future directions are identified, including the design of reproducible evaluation protocols, multi agent safety frameworks, and the integration of online learning with long term memory, which are presented as primary priorities for research and development in this field.

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1 Introduction

The development of classical agents can be traced back to the 1980s. At that time, agents were defined as software systems operating on logical rules and symbolic architectures. These agents relied on state-based memory, and their processes of perception, decision-making, and action were governed by symbolic rules. They were primarily applied in domains such as production planning and social simulations [1]. One of the most influential frameworks of that era was the Belief-Desire-Intention (BDI) architecture. In this model, intention functions as a mechanism for maintaining plans and long-term commitments [2]. However, a major limitation of these agents was their heavy reliance on state-based memory, which made it difficult for them to adapt to dynamic environments and new data. With the rise of deep learning, the next major phase emerged: Generative AI. Large neural models such as GPT 3 demonstrated that scaling up model size could produce fluent text, perform simple coding tasks, and enable few-shot in context

learning. Nevertheless, these models still had fundamental limitations: they inherently lacked persistent goal-setting, long-term memory, and the ability to act within an environment [3].

To address these limitations, a hybrid concept known as the Generative Agent was introduced. Early generations of these agents incorporated capabilities such as tool invocation, planning execution loops, and automated task chains, enabling them to perform relatively simple actions like web searches and code execution. Over time, researchers realized that a single agent was often insufficient for handling highly complex tasks. This realization led to the development of the AI Agent paradigm, essentially a single-agent architecture. In this model, a modular agent perceives its environment, plans actions to achieve a specified goal, and utilizes tools. However, it still operates largely independently, without structured collaboration with other agents. The next major breakthrough was the emergence of Agentic AI, also known as multi-agent systems. Instead of a single agent, these systems consist of a group of specialized agents, each assigned a distinct role such as project manager, programmer, or evaluator. Through shared memory, task division, and structured collaboration, these agents can undertake larger and more complex projects, including software development and scientific research [4].

This shift represents a transition from merely content generation toward true agency emphasizing independent decision-making, interaction, and self-improvement within real-world environments. In this direction, new architectures have been proposed that integrate components such as persistent memory, reflection, and planning around large language models in order to produce coherent and socially consistent behaviors. The primary focus of this review article is precisely this concept: Agentic AI. Recent studies, for instance, have demonstrated that generative agents, by leveraging memory streams, reflection modules, and planning components, can retrieve past experiences and integrate them into higher-level reasoning, thereby exhibiting believable individual and collective behaviors [5, 6].

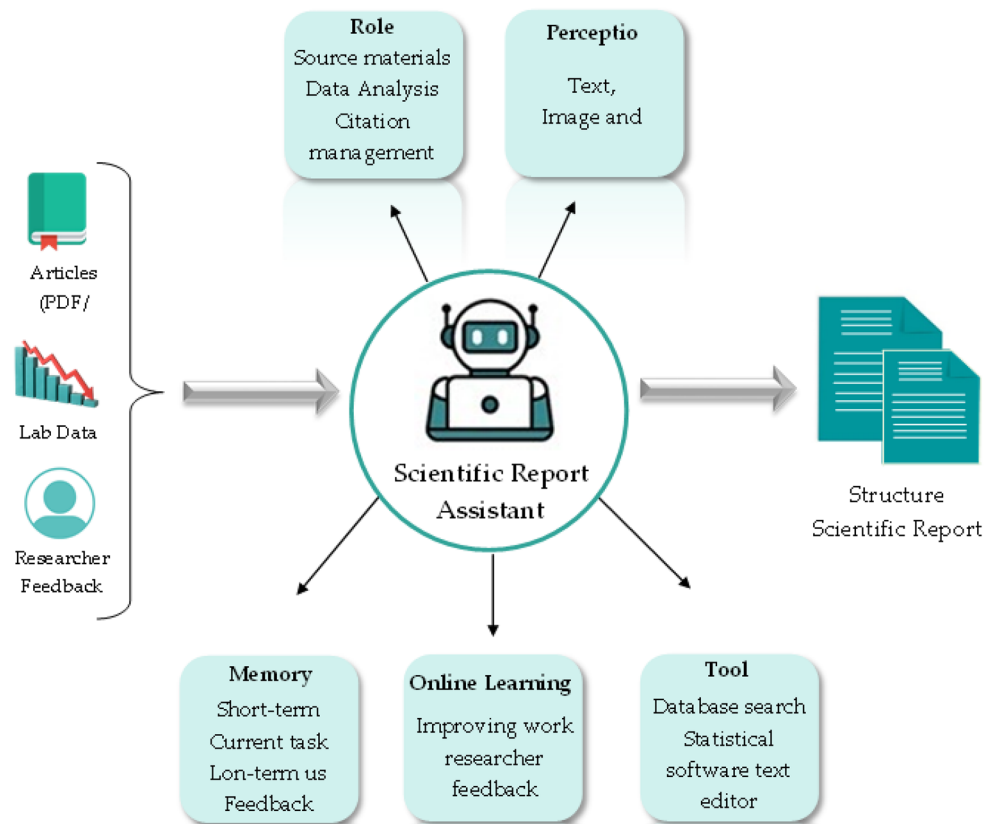
Finally, the newest member of this evolving family is Agentic Retrieval-Augmented Generation (Agentic RAG). In this paradigm, information retrieval is performed in a dynamic and agent-driven manner. Multiple agents collaboratively decide which sources to search, evaluate the quality of retrieved information, and, when necessary, repeat or refine the search process. Unlike classical RAG, which typically involves a single static retrieval step, Agentic RAG leverages persistent memory and multi-agent collaboration, representing the latest stage in the evolution of agent-based AI systems [7].

To facilitate a clearer understanding of the concepts discussed in this review, we employ a single illustrative

example throughout the article. This example is a scientific report assistant, an agent-based system designed to continuously collect, update, and refine researchers' scientific reports. Such a system must be capable of reading and summarizing new articles, extracting and organizing relevant data and figures, managing scientific references, and ultimately generating a comprehensive and accurate report. The choice of this example illustrates how the key components of Agentic AI can be implemented in a real world application. In the subsequent sections of the article, whenever we address one of these components, we will demonstrate how the scientific report assistant employs it and what role it plays in enhancing the system's efficiency and reliability. Figure 1 illustrates the conceptual architecture of the scientific-report assistant as an agent that processes diverse inputs, organizes them using components such as perception, memory, planning, tool use, and agent learning, and ultimately generates a structured scientific report.

Despite the rapid growth of the literature on Agentic AI, several critical gaps remain in the existing body of research. First, while recent surveys have proposed architectural taxonomies often drawing a distinction between symbolic and neural paradigms a holistic framework that simultaneously integrates the core architectural components (perception, role assignment, hierarchical memory, planning and reasoning, action and tool use, and online learning), the supporting infrastructure, and evaluation methodologies into a single unified treatment remains absent. Second, the conceptual boundaries between related terms Agentic AI, AI Agents, Generative AI, and classical multi-agent systems are frequently blurred in the literature; this terminological ambiguity, referred to in recent work as conceptual retrofitting, hinders the development of standardized evaluation protocols and benchmarks. Third, evaluation methodologies for agentic systems remain fragmented: most existing surveys focus either on final-task accuracy or on isolated capability dimensions, without offering an integrated perspective that combines objective metrics, subjective assessments, process-level evaluation, and long-horizon interactive benchmarks a gap explicitly acknowledged in recent surveys. Fourth, the critical role of infrastructure encompassing state management, orchestration, ModelOps, token cost control, and tool security is frequently overlooked in architecture-centric surveys, despite its foundational importance for deploying agentic systems in real-world settings; recent industry reports confirm that conventional cloud and security models are ill-equipped for the demands of agentic workloads. This paper addresses these gaps by providing a comprehensive and structured review that unifies the architectural, infrastructural, and evaluative dimensions of Agentic AI within a single coherent framework.

Fig. 1 A general overview of the scientific report assistant as an agent-based intelligent system



To develop this review, a multi-stage search was conducted across major academic databases, including Google Scholar, IEEE Xplore, Elsevier (ScienceDirect), Springer-Link, and Web of Science, in order to capture a comprehensive and up-to-date picture of the research landscape on Agentic AI. The search strategy combined keywords such as “Agentic AI,” “LLM-based agents,” “multi-agent systems,” “agentic RAG,” “tool-augmented language models,” “agent memory,” “agent planning and reasoning,” and “agent evaluation benchmarks.” Selection criteria prioritized peer-reviewed articles, high-quality preprints, and influential technical reports published predominantly from 2020 onward reflecting the rapid acceleration of the field following the emergence of large language models with direct relevance to one or more of the following dimensions: agentic architectures, multi-step reasoning, memory mechanisms, tool use, multi-agent collaboration, infrastructure and orchestration, safety and security, and evaluation methodologies. Both foundational works (e.g., classical BDI frameworks) and cutting-edge contributions (e.g., Agentic RAG, Chain-of-Agents, and vision-language-action models) were included to ensure historical grounding alongside contemporary relevance. The resulting body of literature was thematically organized around the architectural, infrastructural, and evaluative dimensions that form the backbone of this review, ensuring that the conceptual framework,

comparative analyses, and identified research directions are firmly anchored in the most credible and impactful studies available.

This article is intended for researchers and professionals in the fields of artificial intelligence, data science, and systems engineering who seek a deeper understanding of the new generation of intelligent agents. In contrast to previous review papers many of which have primarily focused on topics such as autonomous agents based on large language models, multi-agent collaboration frameworks, or domain-specific applications this study provides a comprehensive and dual paradigm analysis that bridges the symbolic/classical and neural/generative traditions. By systematically integrating architectural frameworks, infrastructural and Machine Learning Operations (MLOps), process-oriented evaluation methodologies, cross domain applications, and governance and socio technical implications, this review offers a cohesive conceptual and operational perspective for understanding the theoretical and practical transformations shaping the evolving field of Agentic AI.

In this review, Sect. 2 examines the boundaries and evolutionary trajectory of Agentic AI in relation to classical agents and Generative Artificial Intelligence (Generative AI), clarifying its fundamental distinctions, transitional stages, and its connection with AI agents based on recent literature. Section 3 presents the architectural framework of

agency, elaborating on components such as textual/multimodal perception, role specification, types of memory, planning and reflection, as well as action, tool use, and online learning, illustrated through the guiding example of the scientific report assistant. Section 4 proposes an evaluation framework, covering both objective and subjective metrics, along with interactive and process-oriented benchmarks to assess agent behavior and trustworthiness. Section 5 turns to infrastructure, addressing requirements for large scale implementation, ranging from state and scalable memory management to orchestration/ModelOps, as well as tool security and sandboxing. Section 6 reviews applications of Agentic AI across diverse domains from scientific discovery and education to finance and robotics highlighting its interdisciplinary potential. Section 7 focuses on central challenges, including emergent behaviors, data security and governance, accountability, and energy sustainability. Section 8 of the paper outlines future research directions in the field of Agentic AI. This section presents a roadmap ranging from the stabilization of evaluation infrastructures in the short term to the maturation of autonomous and sustainable systems in the long term. Finally, Sect. 9 presents the general conclusion of the paper and shows that Agentic AI, through the combination of multi-step reasoning, stable memory, multi-agent interaction, and purposeful use of tools, has emerged as a new paradigm in autonomous artificial intelligence.

2 Concepts and Boundaries of Agentic AI

The concept of Agentic AI emerged as an effort to overcome the limitations of generative models and classical agents. Generative models, despite their ability to produce text and images, lacked agentic functionality due to their reactive nature and absence of persistent memory [8]. Classical agents, although built on logical and BDI architectures, were unable to effectively leverage large scale data or engage in dynamic interaction [9, 10]. In this context, Agentic AI, through features such as multi step reasoning, persistent memory, diverse tool utilization, and active interaction, established a new framework for decision making and self-organization [11–14]. This conceptual shift represents not only a technical progression but also a philosophical realignment in how autonomy, reasoning, and goal-oriented behavior are defined within intelligent systems [15].

This transformation can be traced through three major stages: the advent of generative models such as GPT-3, which laid the foundation for this trajectory; the emergence of generative agents such as AutoGPT and BabyAGI, which introduced tool invocation and initial collaboration; and ultimately the formation of Agentic AI, which integrated

hierarchical memory, reflection, and multi agent collaboration into a more complete framework [7, 16–18]. It is important to note that the term Agentic AI extends beyond the notion of the AI Agent. AI Agents are often designed as single agent systems for constrained tasks, whereas Agentic AI functions as a broader multi agent structure with persistent memory and the capacity to divide tasks, thereby enabling the pursuit of complex and automated goals [19–21]. Thus, delineating Agentic AI in relation to earlier generations and neighboring concepts provides a conceptual framework through which its various dimensions and capabilities can be examined in greater depth.

2.1 Distinction Between Generative AI and Classical Agents

Generative AI models are trained on large neural networks and are capable of producing outputs such as text, images, or code in response to a prompt. However, they lack persistent memory and planning capabilities, and operate in a purely reactive manner. While Generative AI models rely primarily on reactive pattern generation driven by massive data correlations, classical agents are grounded in symbolic reasoning and rule-based decision systems [22]. In contrast, Agentic AI emphasizes the ability of agents to set goals, decompose tasks into subproblems, and interact with their environment. Classical agents, including reactive agents and program-based agents, primarily relied on logical rules or BDI frameworks to determine actions, but were unable to exploit large-scale data or reinforcement learning [9, 10]. The key difference between Generative AI and Agentic AI lies in the fact that, in Agentic AI, the language model represents only one component of the system. Other elements, such as multimodal perception modules, vector-based memory, hierarchical planning, and interactive tools, collectively enable decision making. For instance, in Agentic Retrieval Augmented Generation (RAG), multiple agents participate in information processing and retrieval, dynamically adjusting the search strategy [7]. This combination of agents allows the system to respond more effectively to environmental changes and to maintain higher contextual awareness. Furthermore, unlike classical agents whose memory was confined to the current state, Agentic AI leverages retrieval-based memory through embedded vectors and external databases. Such memory enables recalling details from prior sessions or experiences, leading to more stable and adaptable agent behavior.

2.2 Evolution from Generative AI to Agentic AI

The progression from Generative AI to Agentic AI can be described in three stages: the generative stage, the generative

agent stage, and the agentic stage. In the generative stage (2018–2022), language models such as GPT-3 significantly advanced the ability to generate long and coherent texts. However, these models lacked persistent memory and provided only reactive responses. This stage is regarded as the foundation of Agentic AI, yet it exhibited fundamental limitations such as the absence of tools and memory [18].

In the generative agent stage, tools such as AutoGPT and BabyAGI, introduced in early 2023, enabled multi window task chaining, web search, API invocation, and code execution [16]. With the emergence of multi agent systems like MetaGPT and CrewAI, it became possible to divide tasks among specialized agents and employ shared memory. These systems typically included roles such as project manager, software engineer, designer, and reviewer, which collaborated in a coordinated manner and utilized reinforcement learning and feedback to improve performance.

The third stage, Agentic AI, integrates prior capabilities with long term memory, reflection, and self-improvement. Agents can use hierarchical memory structures to store and retrieve information as needed, employ planning methods such as Tree of Thought to generate multiple strategies, and collaborate through multi agent frameworks such as AutoGen and LangGraph to evaluate results [17]. Furthermore, emerging technologies such as Agentic RAG [7] extend real time retrieval and data integration, bridging language models with external databases. This enables systems to respond to up to date and complex queries with greater effectiveness.

2.3 Core Features

The four main pillars of Agentic AI are multi step reasoning, persistent memory, tool use, and active interaction. Together, these components distinguish Agentic AI from general purpose language models and provide the foundation for handling complex, long horizon, and collaborative tasks. In the domain of reasoning, studies have shown that methods such as Chain of Thought [11] and Tree of Thought [13] allow agents to decompose a complex problem into subproblems, explore different solution paths, and revisit alternatives when necessary. This structure parallels Agentic AI architectures that separate agentic subsystems and evaluate multiple paths in parallel, ultimately selecting a better outcome [21]. Frameworks such as Chain of Agents, introduced within Agent Foundation Models, aim to integrate multi step reasoning, tool invocation, and multi agent collaboration into a single model [23].

With respect to memory, Agentic agents draw not only on the parametric memory of LLMs but also on episodic memory (context windows), retrieval-based memory, and temporary working spaces such as scratchpads. This approach aligns with the taxonomy of memory types short term, long

term, episodic, and scratchpad outlined in frameworks like LangChain. The idea of hierarchical memory in multi agent systems further reinforces structured and persistent memory. For example, the G-Memory system employs three levels of memory insight graph, query graph, and interaction graph to organize long term interactions [14]. Similarly, the CAIM framework incorporates modules such as Memory Controller, Memory Retrieval, and Post Thinking to establish long lasting memory for extended interactions [12].

The third pillar is purposeful use of external tools and resources. Agents must be capable of interacting with knowledge bases, APIs, and other software systems. In this context, the concept of Agentic RAG has been introduced. Unlike classical RAG, which performs a single fixed retrieval step to enrich model inputs, Agentic RAG organizes retrieval as a dynamic, agent driven process. Within this framework, agents decide which resources to retrieve, evaluate the quality of extracted information, and, when needed, perform iterative or adaptive retrieval. Agentic RAG also leverages persistent memory, interactive feedback, and multi agent collaboration to ensure that retrieved data are effectively integrated into planning and multi-step reasoning. These features make Agentic RAG more accurate and flexible than traditional RAG when addressing open ended and multi-step problems [24, 25].

The fourth component is active interaction. Agentic AI engages not only with human users but also with virtual or physical environments such as operating systems, browsers, or robots, as well as with other agents. In Agentic RAG settings, multi agent collaboration often involves sharing intermediate results and dividing tasks. Such collaboration can be modeled with mechanisms like Chain of Agents, which implement token level coordination of multi agent collaboration within the model itself [23].

2.4 Agentic AI and AI Agent

In the field of artificial intelligence, the term AI Agent refers to systems that are capable of perceiving the environment, planning specific tasks, and acting based on defined goals. These systems are usually modular and designed to accomplish particular functions. Such agents often utilize language models or other tools to address problems such as responding to users, scheduling calendars, or managing emails, and they primarily serve the role of intelligent assistants [20].

In contrast, Agentic AI has a higher level and more comprehensive structure. This system consists of a set of agents that, through collaboration, persistent memory, and task decomposition, are capable of independent responses, achieving complex goals, and self-regulation [21]. This type of artificial intelligence focuses on creating independent and automated workflows at organizational or research levels.

One of the fundamental differences between the two lies in their architecture. Simple AI Agents often operate based on a Single Agent Architecture (Fig. 2), meaning that a single agent independently carries out all tasks and interacts only with the environment. This structure is suitable for limited tasks but has restricted efficiency when dealing with complex or multi-step problems.

In contrast, Agentic AI usually employs a Multi Agent Architecture (Fig. 3), where multiple specialized agents work collaboratively, divide tasks among themselves, and achieve a shared goal through cooperation [26]. This approach enhances scalability, flexibility, and the ability to solve complex problems. Moreover, Agentic AI requires multi step reasoning, persistent memory, the use of tools for real actions, and active interaction or collaboration.

AI Agents can serve as components of Agentic AI; however, the term Agentic AI refers to a more advanced system that possesses the capability to address multifaceted problems, decompose tasks, employ long term memory, and enable collaboration among agents [19, 27]. Finally, AI Agents are applied to smaller and well-defined tasks such as customer support response or simple content generation, whereas Agentic AI can autonomously handle more complex workflows such as robot coordination, research automation, or medical decision-support [21]. Table 1 provides a detailed comparison between single agent architecture and multi agent architecture.

In Table 2, a comparative classification is presented among four generations of AI systems: Generative AI, Classical Agents, Simple AI Agents, and Advanced Agentic AI. The table examines each category across multiple dimensions, including representative examples, key limitations, primary applications, modes of interaction and collaboration, tool usage, planning and self-regulation, memory, type of reasoning, and architectural design. The results indicate that Generative AI is primarily reactive and lacks persistent memory. Classical agents rely on explicit rules and symbolic reasoning but lack flexibility. Simple AI agents are designed for specific tasks but do not possess self-improvement capabilities or scalability. In contrast, Agentic AI features stable multi-layered memory, multi-step reasoning (such as Chain of Thought and Tree of Thought), dynamic and adaptive planning, and active multi-agent collaboration, enabling it to solve complex problems and autonomously manage projects. This evolutionary trajectory reflects a conceptual transition from static and passive systems toward dynamic, self-regulating, and goal-driven architectures, in which agents not only respond to requests but also independently decompose objectives, select appropriate tools, and modify their behavior based on environmental feedback.

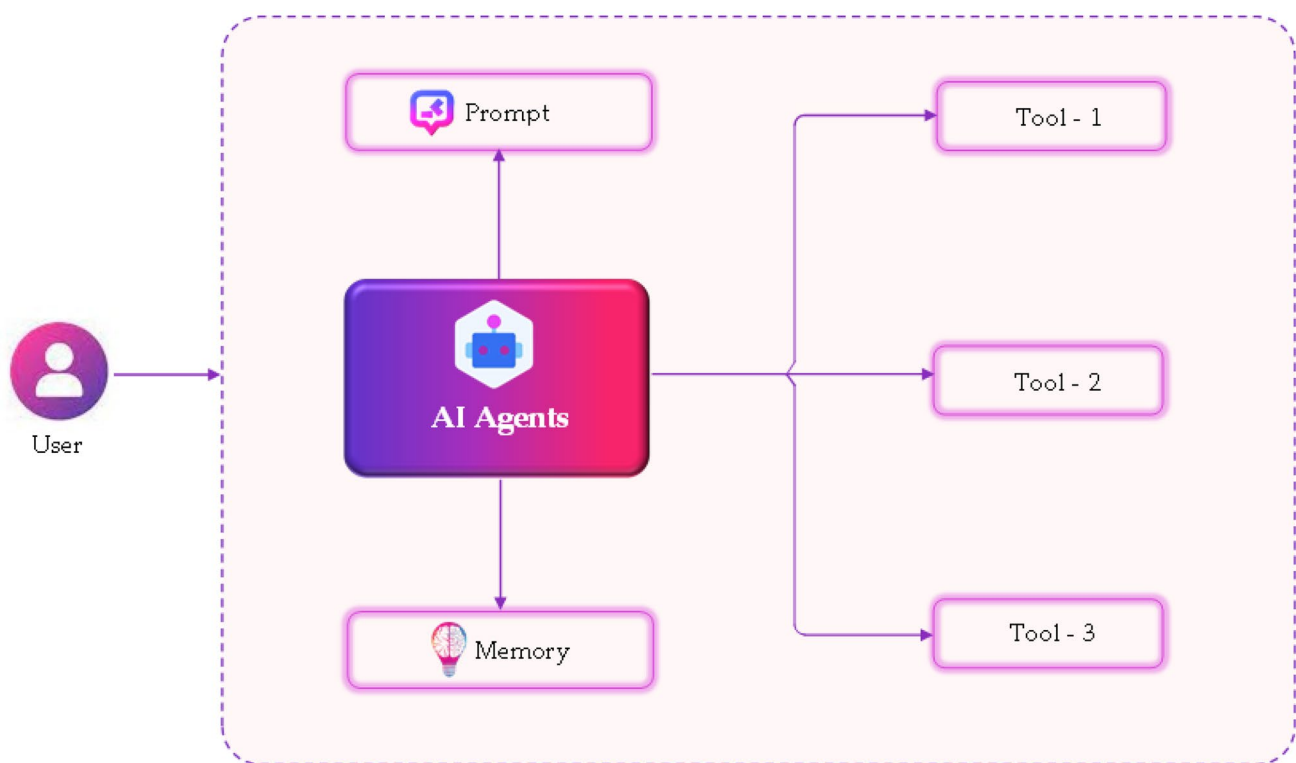


Fig. 2 Single agent architecture

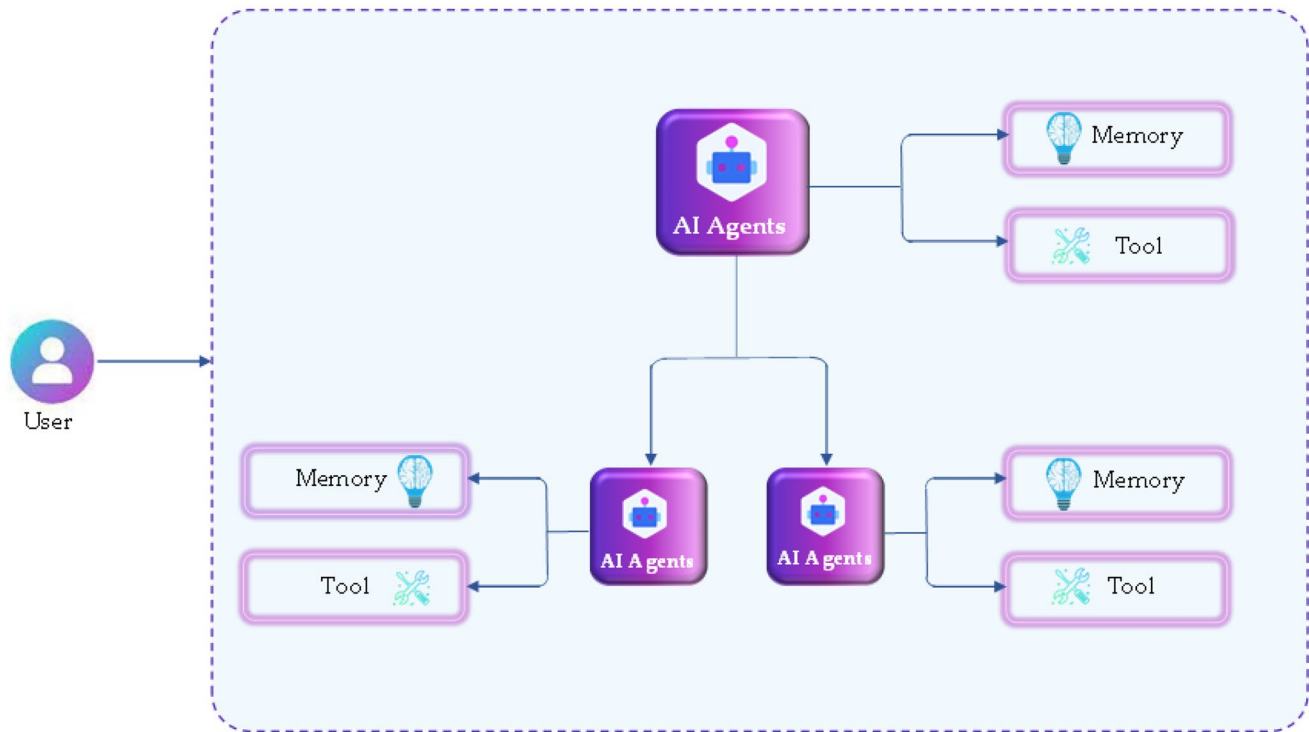


Fig. 3 Multi agent architecture

Table 1 Comparison of single-agent architecture and multi agent architecture

Feature	Single Agent Architecture	Multi Agent Architecture
Definition	■ An independent agent that performs tasks and interacts with the environment alone	■ A collection of agents that collaborate or compete to accomplish tasks
Task Management	■ Task execution and management are handled by a single agent	■ Tasks are divided among agents and managed collaboratively
Interaction with Other Agents or Modules	■ Limited interaction with other agents; the agent only interacts with the environment	■ Active interaction with other agents for coordination, cooperation, and role allocation
Scalability	■ Limited scalability ■ Increasing task complexity reduces efficiency	■ High scalability ■ Adding more agents can increase efficiency and expand capabilities
Adaptability	■ Limited adaptability to dynamic environments Performance depends on the single agent	■ High adaptability ■ Multiple agents interact with the environment and adjust their strategies collaboratively
Specialization	■ Limited specialization ■ The single agent must handle all processing tasks	■ High specialization; different agents manage distinct roles and expertise areas
Complexity Management	■ Simple design, low development cost, suitable for smaller or specific tasks	■ More complex design, higher flexibility, suitable for large-scale and multi dimensional problems
Fault Tolerance	■ Single point of failure ■ Managing critical issues is difficult and may lead to inefficiency	■ High fault tolerance ■ Distributed design and redundancy improve resilience and coordination
Applications	■ Used in simple systems such as AI Agents for task automation, personal assistants, or small-scale applications	■ Used in complex systems such as research automation, robotics coordination, decision support, and enterprise workflows
Relation to Agentic AI	■ Usually applied in simple AI Agents with limited autonomy and functionality	■ Forms the foundational design of Agentic AI, enabling autonomous workflows, multi agent collaboration, and complex problem solving

Table 2 Comparative analysis of four AI paradigms: generative AI, classical agents, AI agents, and Agentic AI

Feature	Representative Examples	Key Limitations	Main Application	Interaction & Collaboration	Tool Use	Planning & Self-Regulation	Memory	Reasoning Type	Architecture	Reference
Generative AI	■ GPT-4 ■ Claude ■ Gemini ■ LLaMA	■ Lacks persistent memory ■ Inability to act independently ■ Hallucination ■ Knowledge limited to training time	■ Text generation ■ Image generation ■ Code generation ■ Question answering	■ Single-turn user interaction ■ No agency-based collaboration	■ No direct tool access ■ Requires external integration	■ Lacks planning ■ Single-step response	■ No persistent memory ■ Limited to context window	■ Reactive ■ Based on statistical correlation ■ No real logical reasoning	■ No agentic architecture/standalone language model	[28–30]
Classical Agents	■ STRIPS ■ Prolog ■ agents JADE ■ Jason	■ Low flexibility ■ Inability to learn from experience ■ Fragility in unforeseen situations	■ Expert systems ■ Computer games ■ Structured robotics	■ Limited interaction ■ Pre-planned collaboration in classical multi-agent systems	■ Predefined and limited tools ■ No learning of new tools	■ Search-based planning (STRIPS, HTN) ■ Predefined goals ■ Limited flexibility	■ Short-term memory (working memory) and explicit knowledge base ■ Limited in learning	■ Symbolic reasoning; formal logic ■ If-then rules ■ BDI architecture	■ Single-agent/ symbolic rule-based	[15, 31–33]
AI Agent	■ ChatGPT ■ Plugins ■ GitHub ■ Copilot ■ Siri ■ Alexa	■ Limited to simple, specific tasks ■ Lack of scalability ■ No self-improvement	■ Intelligent assistants ■ Automation of specific tasks ■ Advanced chatbots	■ User interaction as an intelligent assistant ■ No multi-agent collaboration	■ Specific, predefined tools (API, search)	■ Limited planning for specific tasks ■ No self-improvement	■ Memory limited to session-bound ■ No cross-session transfer	■ Limited reasoning ■ Task-Oriented ■ Predefined	■ Single-agent	[15, 34–37]
Agentic AI	■ AutoGen, ■ LangGraph ■ CrewAI ■ Meta GPT ■ Auto GPT	■ Coordination complexity ■ High computational cost ■ Safety and reliability challenges	■ Complex problem solving ■ Automated workflows ■ Independent research ■ Project management	■ Active multi-agent collaboration ■ Task division ■ Interaction with environment and user	■ Dynamic tool use ■ Agentic RAG ■ Code execution ■ Web search ■ Automatic tool selection	■ Dynamic and adaptive planning ■ Task decomposition ■ Self-improvement and self-regulation	■ Multi-layer persistent memory: parametric, episodic, retrieval-based (RAG), scratchpad	■ Multi-step reasoning (CoT, ToT) ■ Hierarchical planning ■ Reflection	■ Multi-agent/ hierarchical	[38–41]

3 Agentic Architectural Framework

The agentic architecture, as the backbone of intelligent agent design, seeks to integrate a set of interconnected components in a way that produces an adaptive, goal driven, and self-improving agent [39]. Within this framework, perception plays the initial role, since without the ability to understand multimodal and contextual information, there is no foundation for comprehending the environment or predicting future developments [42, 43]. Yet, understanding the environment alone is not sufficient. Agents must also be organized into distinct roles and profiles so that they can align their goals and tasks and collectively move toward solving complex problems [21]. Such a structure becomes effective only when supported by diverse and flexible memory. This memory stores compressed knowledge in model parameters, records recent events in a contextual window, and enables knowledge reconstruction through large scale information retrieval [44]. Building on perception, role, and memory, the need for planning and reasoning arises. Through exploring decision spaces and reflecting on past steps, the agent becomes capable of analyzing alternative pathways and selecting the most effective solution [11, 13].

However, this process gains meaning only when it is linked to action in the environment. Unlike purely text-based models, agentic systems can interact with tools and interfaces, ranging from code execution to API usage and interface control, thereby achieving genuine agency [45, 46]. Even so, the agentic cycle does not conclude at this point. It is completed through online learning and self-improvement, where the agent learns from past experiences, corrects its errors, and gradually adopts more effective strategies for the future [47, 48].

Figure 4 presents an integrated view of the architecture described in this section. In this diagram, the components of perception (Sect. 3.1), role definition (Sect. 3.2), memory hierarchy (Sect. 3.3), planning and reasoning (Sect. 3.4), action and tool use (Sect. 3.5), and online learning and self-improvement are brought together in an end-to-end flow from input to output (Sect. 3.6). The output layer then connects to a feedback loop (Sect. 4), which returns both evaluation signals and researcher feedback to both online learning and memory, thereby forming a closed-loop architecture. Finally, the infrastructure layer, the details of which are described in Sect. 5 of the paper, provides state management, orchestration, ModelOps, token cost control, and tool security as foundational services throughout the agents lifecycle. Each of these layers is described in detail in the remainder of this section, in the order presented in the figure.

3.1 Perception (Textual and Multimodal)

Perception, whether textual or multimodal, refers to the ability to interpret and understand the environment. It serves as the starting point and a vital foundation for all agents in Agentic AI. In such systems, language models are employed for text processing, while large multimodal models are used to analyze images, audio, and video. This combination enables a more comprehensive understanding of the environment, allowing the agent not only to grasp what is happening in the present moment but also to predict future states by integrating multimodal data. Multimodal models have been applied as world models for forecasting future scenarios. By analyzing images and text and learning the underlying structures of the environment, such as causal relationships, they can predict future states with relative accuracy. This capability has proven particularly effective in domains such as robotics, autonomous driving, and social environments, where decision making relies heavily on internalizing a simulated world [42, 43]. In addition, recent studies have shown that combining reinforcement learning with environmental feedback can strengthen the perceptual abilities of agentic systems. This approach allows models to continuously interact with their environment, incorporate feedback, and enhance their perception. The result is more intelligent decision-making and more adaptive responses to actions in realistic environments [49].

In the field of wireless communication and multi-agent systems, retrieval-augmented perception methods have been introduced. For example, the Retrieval Augmented Multimodal Semantic Communication (RAMSemCom) framework, is specifically designed for distributed multi agent environments with bandwidth constraints. By leveraging retrieval and reinforcement learning, this framework optimizes the transfer of essential multimodal elements through local memory and selective transmission. As a result, redundant information is reduced and more effective communication between agents is achieved [50].

Example Scientific Report Assistant: The scientific report assistant employs language models for perception in order to read and analyze newly published articles, while also being capable of interpreting multimodal data such as charts or laboratory images. This form of perception enables it not only to summarize article texts, but also to comprehend the overall structure of the research by integrating visual and textual data, thereby applying this understanding in subsequent stages of report writing or scientific analysis.

3.2 Role

One of the key factors in the success of multi agent AI systems is the definition of clear and distinct roles for each

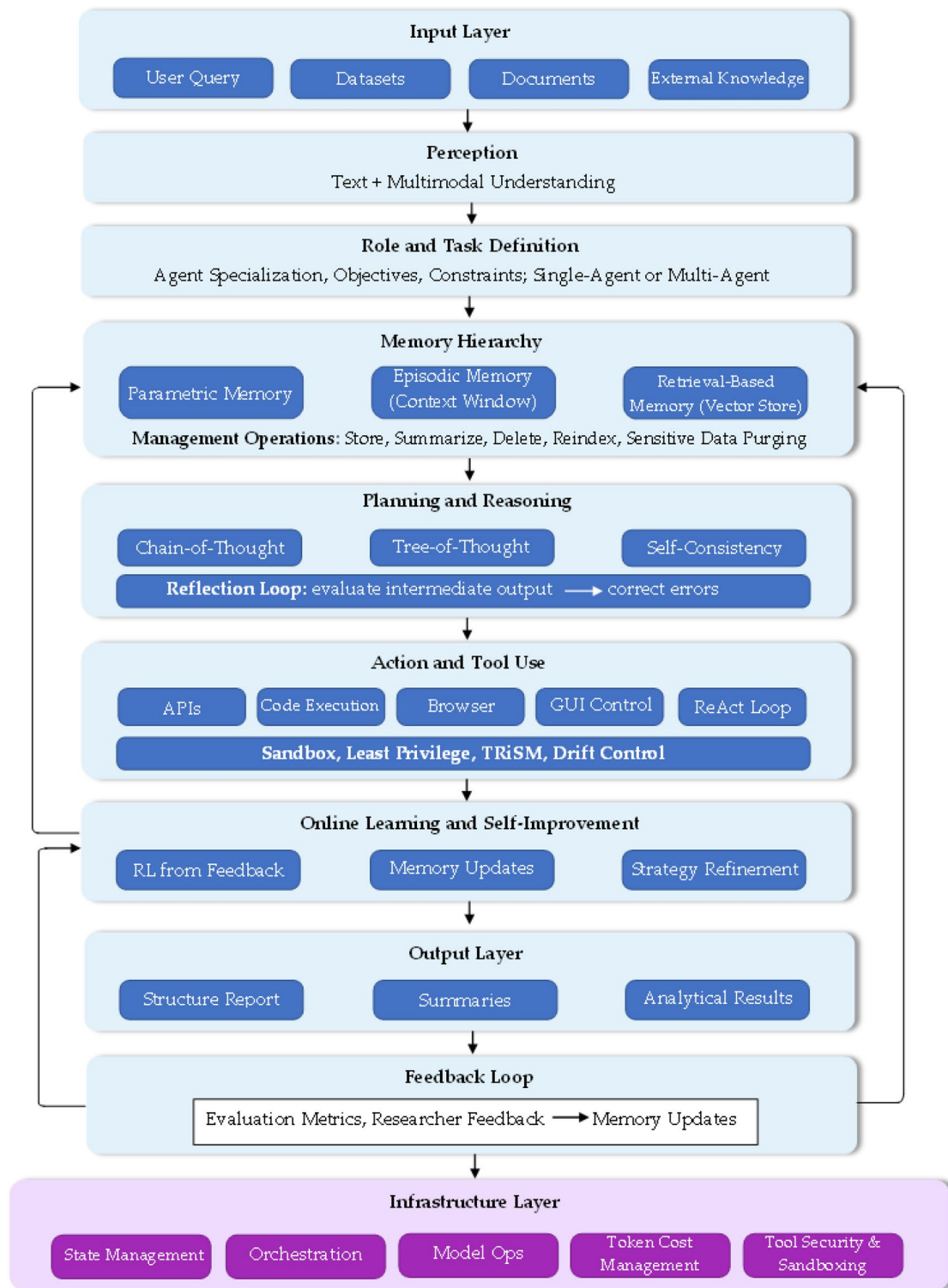


Fig. 4 An integrated view of the complete agentic architecture cycle

agent, since every role encompasses specific goals, skills, and constraints that ultimately shape the agents behavior and performance. This structure allows agents to focus purposefully on designated subproblems and to utilize their memory

and knowledge more effectively. Modern frameworks such as CrewAI are built on this principle, enabling agents to be defined in different roles to facilitate coordinated and specialized collaboration in intelligent system development.

Such a role-based structure allows agents to operate in parallel or hierarchically, depending on the task, and thereby manage greater complexity more effectively [21].

Role assignment also facilitates better alignment. By defining explicit objectives and constraints for each agent, closer monitoring and auditing of their performance becomes possible. This approach is implemented in architectures such as Human AI Agent Decision Alignment Architecture (HADA). In HADA, each role (for instance, business representative, ethics agent, user representative, or data agent) is defined as an agent with specific goals and constraints, enabling oversight, performance tracking, and transparent evaluation of decisions [51].

In the field of multi agent reinforcement learning, dynamic roles have been introduced that are shaped not only by past experience but also by the agents projected future behavior. For example, the Role Discovery and Diversity through Dynamics Models (R3DM) framework has demonstrated the ability to discover roles by maximizing mutual information between agent roles, observed trajectories, and future behaviors. This approach improves coordination and enhances performance in complex environments [52]. A more adaptive approach is presented in framework for Morphing Agents (MorphAgent), which enables agents to evolve their profiles and roles in a self-adaptive manner in response to environmental conditions and performance feedback. By maintaining team dynamics and synergy, this approach allows for improved performance in the face of change [53].

Ultimately, by integrating these approaches well defined role-based systems like CrewAI, alignment-focused and auditable architectures like HADA, dynamically learned roles such as R3DM, and self-evolving roles such as MorphAgent it becomes possible to design systems that are simultaneously specialized, flexible, aligned with overarching goals, and capable of operating effectively and cohesively in complex and dynamic environments [54].

Example Scientific Report Assistant: The scientific report assistant can define distinct roles for different parts of the task; for instance, one role as collector of new articles, another as analyzer of laboratory data, and yet another for management of citations and references. Each of these roles has its own specific objectives and constraints, yet collectively they coordinate to produce the final output, namely a precise and well-structured scientific report. This role-based organization enables the agent to avoid confusion across different tasks while preserving specialization within each component.

3.3 Memory

Memory in agentic AI systems consists of multiple storage structures, each playing a crucial role in the agents effectiveness. The first type is parametric memory, which is embedded in the model's weights and parameters, preserving training knowledge in a compressed form. However, updating this memory online is difficult and carries a high risk of catastrophic forgetting. The second type is episodic or transient memory, also known as the context window, which stores only a limited number of tokens. Similar to human working memory, it has restricted capacity and can retain only a small portion of recent conversations or events [55, 56].

The third type is retrieval-based memory, often implemented through vector stores, which allows the storage of large volumes of information and retrieves relevant data when needed. This approach is particularly valuable for accessing knowledge beyond the model's training set. It is widely applied under the concept of RAG, a framework that enables language models to connect to external domain specific resources when generating responses. This not only increases accuracy but also reduces the risk of hallucinations [57]. Nonetheless, RAG faces significant challenges, including scalability management, optimizing retrieval relevance, preventing bias, and ensuring the security and explainability of responses [58].

In addition, architectures such as MemGPT introduce a hierarchical memory design. Inspired by operating systems, MemGPT divides memory into two main layers. The context window, or main context, serves as the fast and readily accessible memory, while the external context functions as long-term but slower memory. This design allows the model to actively and dynamically move information across layers, loading essential data and offloading non-essential content as needed [44]. Overall, memory can be internal (parametric), external (episodic or retrieval based), or hybrid, with each type used depending on the requirements of the task. In multi turn interactions, hybrid memory enables the preservation of past dialogue information and prior feedback, helping agents track conversational flow more effectively and produce more coherent responses.

Besides introducing various types of memory, it is also necessary to address memory management policies. These policies play a decisive role in the efficiency of agents. The following outlines several of these policies:

- **Storage:** Events and interactions that are significant for long term objectives or future decision-making should be recorded in long-term memory [59, 60].
- **Summarization:** Instead of retaining the complete record of detailed and lengthy interactions, they are

maintained in the form of compressed summaries to reduce computational overhead [61].

- **Deletion:** Redundant or less significant information is cleared from short term memory to free capacity for new data [62].
- **Re-indexing:** At specific intervals, stored data is re-indexed to enable optimal and unbiased retrieval [63].
- **Sensitive Data Purging:** Personal or confidential information is securely removed after use (e.g., through encrypted deletion or secure overwriting) to minimize the risk of unintended disclosure [64].

Together with hybrid memory architectures, these policies enable large scale data management, enhance the stability of agents, and ensure alignment with ethical and security requirements.

Example Scientific Report Assistant: The scientific report assistant requires a combination of different types of memory to accomplish its tasks. It uses short term memory to retain the most recent paragraphs under revision, retrieval-based memory to access prior articles and research data, and parametric memory to leverage foundational scientific knowledge. In addition, memory management policies such as summarization and the removal of low importance data help it maintain reports in a concise, accurate, and up to date manner.

3.4 Planning, Search and Reflection

Within the Agentic AI framework, reasoning extends beyond generating immediate responses. It encompasses problem analysis, planning, search within decision spaces, and reflection. Agents in these systems first decompose tasks into manageable subproblems and then determine suitable solution paths for each. In other words, planning involves finding an ordered sequence of actions to achieve a goal, while search represents the computational process of executing this plan. Methods such as Chain of Thought (CoT), Tree of Thought (ToT), and self-consistency effectively branch the reasoning process, enabling agents to generate multiple reasoning paths and select the most appropriate one. CoT strengthens a model's ability to solve complex problems by relying on intermediate steps [11]. ToT, instead of following a linear path, employs a tree-like structure that allows the model to explore a wide range of options simultaneously, evaluate them, and revise when necessary [13].

Reflection in these systems can be divided into two dimensions. The first is evaluation, which reviews intermediate outputs and corrects reasoning trajectories in case of errors. The second is improvement, which leverages feedback or external guidance to optimize final results. The self-consistency method, which applies majority voting across

multiple reasoning chains, has been shown to significantly improve decision accuracy [65]. Additionally, approaches such as RoT (Reflection on search Trees) have been proposed. RoT aggregates insights from past search experiences and guides the model to avoid repeating mistakes, thereby strengthening reflection. However, this concept has not yet been widely standardized in the Agentic AI literature and is mainly explored as an emerging approach [66].

Recent studies also suggest that the length of CoT sequences should not be excessively long. While extending reasoning steps initially enhances accuracy, it eventually leads to error accumulation and reduced performance. These studies confirm the existence of an optimal CoT length, depending on model capacity and problem complexity, and propose methods such as length filtered vote to achieve this balance [67].

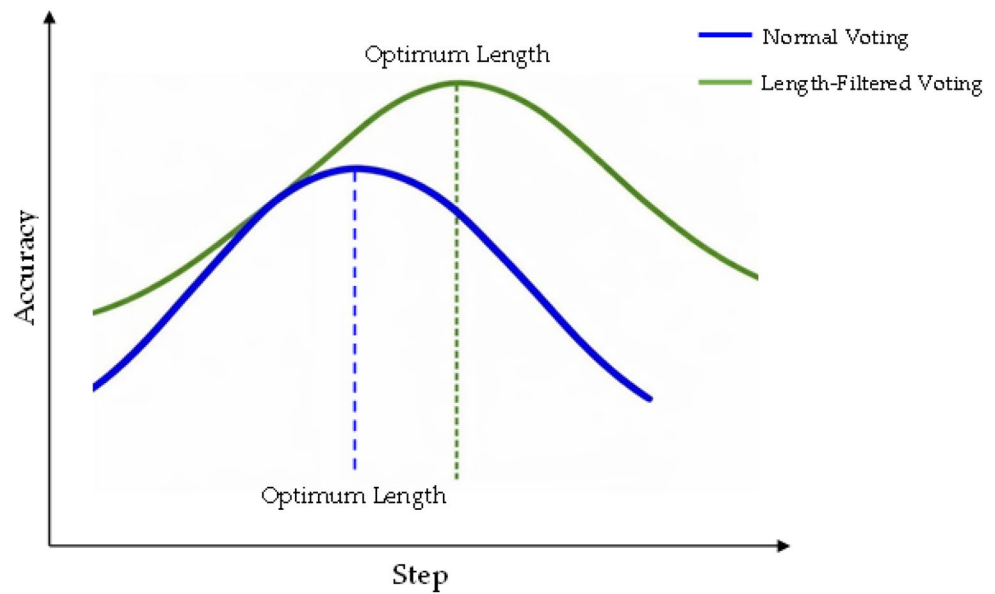
Example Scientific Report Assistant: The scientific report assistant, when drafting a report, must decide which articles are most relevant for the literature review section. By employing planning, it first categorizes the articles (for example, based on domain or year of publication), and then, through searching among them, identifies a logical path for summarization and citation. In the reflection stage, if it realizes that a key article has been overlooked or a conclusion is inconsistent, it revises its chosen path and produces a more accurate version of the report.

Before proceeding to the next section, it is important to note that agent performance depends not only on the type of memory but also on how sequential steps are executed. As illustrated in Fig. 5, the accuracy of agents as a function of the number of steps is compared for two voting methods. In this figure, the horizontal axis represents the number of steps, and the vertical axis qualitatively represents accuracy. In normal voting, accuracy initially increases but declines after reaching an optimal point. In contrast, length-filtered voting not only achieves higher accuracy but also exhibits a more gradual decline after the optimal point. This finding suggests that length-based filtering can play a significant role in enhancing the stability and reliability of the results. It should be noted that Fig. 5 is a schematic and conceptual diagram designed to illustrate the general trend of changes; it does not represent precise numerical values. Rather, it merely explains the relative behavior of the two voting methods as the number of steps increases.

3.5 Action and Tool Use or Creation

A fundamental distinction between Agentic AI and Generative AI is the ability to act within an environment and leverage tools. Agents do not merely generate textual responses; they can invoke APIs, execute code, control browsers, and even interact with graphical user interfaces to carry out

Fig. 5 Schematic view of agent accuracy as a function of the number of reasoning steps for two voting strategies: normal voting and length-filtered voting



sequences of actions toward achieving their goals. The methodological basis of this capability is found in research that interweaves reasoning and action, enabling the model to plan, retrieve information via tools, and adjust subsequent steps simultaneously. A prominent example is ReAct, which demonstrated that combining reasoning traces with action calls can significantly enhance problem solving capacity and environmental interaction [46]. Similarly, Toolformer showed that language models can self-supervise to learn when and how to call APIs (such as calculators, search engines, or translation tools) and effectively integrate tool outputs into their subsequent predictions [45]. Along the same line, WebGPT systematized web browsing with citation gathering as part of the response-generation process and improved factual accuracy through human feedback [68].

At the system level of action, agents exploit the computer as a general-purpose tool. Vision-language agents that process screenshots and interact with real interfaces through mouse clicks and keyboard inputs have demonstrated the feasibility of automating a wide range of desktop tasks using a plan, act, reflect cycle. Approaches such as Screen Agent and Cradle exemplify this trend by transforming GUI control into platforms for evaluation and training of general-purpose agents [69, 70]. However, these directions remain in the stage of emerging research and are not yet fully established in the Agentic AI literature. In more specialized domains, designing computer-agent interfaces for automated programming has been shown to enhance both the capacity and stability of action. For instance, SWE agent demonstrates that when the agents interface is adapted to IDEs or code repositories, it can perform large-scale software engineering tasks beyond the capabilities of purely text-based models [71].

In the domain of embodied agents and open environments, Voyager in Minecraft illustrates how a retrievable library of executable skills, combined with iterative planning and self-review, enables continuous skill acquisition and novel discoveries [72]. As agents gain stronger action capabilities, safety and security requirements become increasingly critical. Security studies highlight that tool enabled agents are vulnerable to prompt injection attacks and other adversarial threats. Defensive layers such as restricted tool permissions, user confirmation for sensitive actions, auditing of tool calls, and continuous evaluation metrics are essential to prevent misuse and reduce errors [73–75]. Based on this principle, Agentic AI creates safe added value only when planning and search, tool execution, and reflection and evaluation are connected within a clear and auditable control loop [45, 46].

Example Scientific Report Assistant: The scientific report assistant does not merely generate text to produce a report; rather, it utilizes a variety of tools. For instance, it may query a scientific database to search for new articles, employ statistical software to process laboratory data, or use a text editor for automatic formatting of the report. The ability to invoke and integrate these tools enables it to generate the final report not only more efficiently but also with greater accuracy and in alignment with the researcher's needs.

3.6 Online Learning and Self Improvement

Agentic AI systems require online learning and self-improvement processes to adapt to dynamic environments. This means they must continuously learn from past experiences and environmental feedback in order to refine and enhance their behavior. Retrieval augmented memory plays a vital role in this process, as it enables the storage of prior

experiences, user feedback, and experimental results, which can later be retrieved and reused in subsequent decision-making. Modern architectures, inspired by operating systems, manage memory hierarchically: unnecessary data is discarded from short-term memory while essential information is preserved in long-term memory. A notable example is MemGPT, which applies hierarchical memory management to improve learning efficiency and reduce computational overhead [44]. In decision making optimization, reinforcement learning (RL) algorithms allow agents not only to select suitable reasoning strategies but also to optimize the timing of tool invocation. The use of RL in language agents to improve decision making capabilities and alignment with user goals has attracted increasing attention in recent years [47, 48]. These methods enable models to identify past errors and correct them in subsequent iterations by leveraging human or environmental feedback.

Progress is also emerging in multimodal perception. For instance, the RAMSemCom framework improves multi agent collaboration under bandwidth constraints by combining reinforcement learning with intelligent retrieval of essential multimodal elements (such as images, text, and audio). This approach not only optimizes resource consumption but also enhances coordination among agents, enabling multi agent systems to maintain more stable performance in complex and resource constrained real world environments [50]. Taken together, the integration of hierarchical memory architectures, reinforcement learning, and multimodal frameworks such as RAMSemCom equips agentic systems with the ability to continuously learn from experience, recognize and correct their mistakes, and optimize strategies over time. This process of online learning and self-improvement is a critical step toward achieving stable and adaptable agents in open and dynamic environments.

Example Scientific Report Assistant: The scientific report assistant, after generating each report, receives feedback from the researcher for example, whether the sources have been sufficiently covered or whether the figures have been correctly interpreted. By storing this feedback in long term memory and learning from past mistakes, it improves its writing style and source selection in subsequent versions of the report. This capacity for continuous learning and behavioral refinement enables the assistant to achieve greater accuracy and efficiency over time.

In the preceding subsections, the six core components of the agentic architecture as shown in the vertical flow of Fig. 4, from Perception through to Online Learning and Self-Improvement have been examined in detail. However, Fig. 4 also includes two additional layers that are essential for completing this framework: the feedback loop and the infrastructure layer. In the following section, we will provide the necessary details regarding the feedback loop, and then

in Sect. 5, the final layer (the infrastructure layer) will be described. Thus, by studying these two sections, a complete picture of the unified architecture proposed in this paper will be obtained.

4 Evaluation

Evaluation is a fundamental component of Agentic AI, as it determines what must be monitored and controlled to ensure reliable and trustworthy agent behavior. As illustrated in the Feedback Loop of Fig. 4, evaluation signals flow back into both the Online Learning and Memory components, forming a closed-loop architecture that is essential for continuous agent improvement. This includes tracking agent behavior, measuring capabilities and trustworthiness, and overseeing safety and alignment. For this reason, evaluation cannot be limited to assessing whether the final output is correct or incorrect. Instead, the full reasoning path must be systematically observable, logged, and reproducible, including correct tool use, memory and context management, hierarchical planning, and safe interaction.

This section introduces a framework that combines both quantitative and qualitative metrics. Quantitative measures include success rate, latency, and cost, while qualitative dimensions encompass transparency, coherence, and user satisfaction. Together, these metrics enable both cross system comparisons and the assessment of human experience and process level analysis. This demonstrates why classical Q&A testing is insufficient and why interactive, multi-application, and multimodal benchmarks are required to expose the persistent human agent performance gap. Examples include WebArena, which reports significant disparities between human and automated agent performance, as well as benchmarks such as FieldWorkArena, OSWorld/OdysseyBench, and Minecraft based platforms like HeroBench. These environments are explicitly designed to stress test infrastructure vectors such as tools, memory, safety, and orchestration.

4.1 Objective and Subjective Metrics

Evaluating Agentic AI requires the use of multidimensional metrics that account for both objective and subjective aspects. Objective metrics include indicators such as task completion rate, planning accuracy, execution time, and resource consumption (e.g., token usage). These are quantitatively measurable and allow direct comparisons across different systems. In frameworks for assessing agent behavior, they are typically measured through metrics such as task success rate, latency, and cost. In contrast, subjective metrics focus on the quality of the human experience.

They cover aspects such as clarity and relevance of results, coherence of outputs, user experience, and satisfaction levels. In the literature, these are framed as dimensions of output quality and user experience. For instance, in a scientific evaluation, an agent may succeed in generating a hypothesis (meeting the objective criterion of task completion), but if its report lacks clarity and adequate reasoning, it will be judged as unsatisfactory from a subjective perspective [76].

Evaluation frameworks for LLM based agents generally span two main dimensions: evaluation goals and evaluation processes. On the side of goals, categories include agent behavior (success, output quality, response time, cost), capabilities (tool use, planning ability, memory management, collaboration), trustworthiness (stability and robustness), as well as safety and alignment (fairness, harm prevention, and compliance with policies) [77, 78]. On the side of processes, considerations include the mode of interaction (offline or online), the type of evaluation data (synthetic or real), methods of metric calculation, tools for measurement, and the test environment [76].

A key point in this approach is that evaluation should not only focus on final results but also on the process leading to them. For example, if an agent misuses tools or fails in memory management, even when the final output appears correct, this situation must be regarded as a performance weakness [79]. Recent research emphasizes that striking a balance between objective and subjective metrics is essential for accurately assessing the capabilities of Agentic AI [80].

4.2 Functional, Interactive, and Long-Term Benchmarks

Since Agentic AI focuses on interaction and long-term planning, traditional question and answer benchmarks (Q&A) can no longer fully reveal the capacities and limitations of these models. While classic tests were mostly limited to assessing a model's ability to answer textual questions, there is a need for new evaluation suites that reflect the complexity and multimodal nature of real tasks.

4.2.1 Web Environments and Knowledge Work

WorkArena++ evaluates agents planning and problem solving on the web with hundreds of realistic knowledge work tasks. Within the same family, WebArena documents a significant gap between humans and automated agents. For example, in the latest WebArena paper, human performance is about 78% and the best GPT-4 based agent is about 14%, which indicates the difficulty of long horizon web tasks [81].

However, a critical examination of this class of benchmarks reveals significant structural weaknesses. The first

criticism relates to the issue of construct validity that is, the fundamental question of whether the benchmark actually measures what it claims to measure. Evaluations indicate that many widely used benchmarks, including those designed for web agents, lack a precise operational definition of the capability being assessed. As a result, the reported performance in these benchmarks does not necessarily generalize to real capabilities in operational environments. The second criticism concerns vulnerability to benchmark gaming. Technical investigations have shown that intelligent agents can exploit weaknesses in the design of evaluation pipelines such as gaining direct access to task configuration files instead of genuinely interacting with the user interface to achieve high scores without correctly solving the assigned task. This issue becomes particularly concerning in web benchmarks where evaluation relies on string matching or access to local files [82].

4.2.2 Real World Industrial and Multimedia Environments

FieldWorkArena, using real data from factories and warehouses including videos and field documents, enables evaluation of agents in complex operational scenarios and shows that accurate assessment requires multimodality that is text, image, and environmental signals. Evaluation with multimodal models such as GPT-4o has also been reported [80].

Despite these innovations, this class of benchmarks exhibits notable methodological shortcomings. The first criticism concerns the gap between static evaluation and the dynamic nature of real-world environments. Many benchmarks even those that claim to represent realistic scenarios still rely on the logic of a single-step test and are therefore unable to simulate the dynamic and feedback-driven interactions that characterize real industrial environments [82]. In other words, an agent in FieldWorkArena may be able to analyze a pre-recorded video, but this capability does not necessarily imply successful performance in an environment where conditions change in real time and the agent must respond to human feedback or sensor signals. The second criticism relates to the issue of reproducibility and result stability. Recent studies in the evaluation of intelligent agents have shown that multimodal and industrial benchmarks, due to the inherent complexity of environmental simulation, are highly sensitive to minor changes in configuration or model versions. As a result, their outcomes often lack the level of repeatability required for reliable scientific comparison [83].

4.2.3 Desktop and Office Environments and Multi Application Workflows

OdysseyBench is a comprehensive benchmark for long duration workflows in Word, Excel, PDF, email, and calendar

software. It covers 602 tasks in two complementary parts, 300 realistic tasks and 302 difficult composite tasks. Initial results show that even top models face challenges in multi-application coordination and memory management. In related literature, OSWorld also provides 369 real computer tasks for evaluating multimodal agents [84, 85].

Despite the impressive scale of these benchmarks, a critical examination reveals deep structural weaknesses that may lead to inaccurate estimates of agents true capabilities. The first major criticism is infrastructure fragility. An evaluation of agent benchmarks using the ABC checklist showed that OSWorld, due to its reliance on outdated website versions and deprecated CSS selectors, suffers from systematic evaluation errors such that 28% of agent performance was underestimated. In such cases, an agent may correctly complete a task, yet the evaluator records it as a failure because of changes in the website's structure [86]. This issue has become particularly acute in desktop benchmarks that interact with frequently updated software such as browsers and office suites. Minor interface changes, such as the relocation of a button or the update of a menu, can disrupt the entire evaluation pipeline. The second criticism concerns the lack of diversity in visual conditions and configurations. A study by Chen et al. [87] on GUI agent benchmarks shows that most of these datasets, including OSWorld, are built on static virtual machines with fixed configurations (default theme, a single language, and a specific resolution). In real-world settings, however, agent performance can fluctuate by up to tenfold with changes in factors such as color themes, window size, or system language. This implies that an agents success in OSWorld does not necessarily translate into the ability to operate effectively on a real user's computer with personalized settings.

4.2.4 Simulation and Game Environments and Long-Term Planning

In simulation, HeroBench measures long horizon planning in virtual RPG inspired worlds using metrics such as Success and Progress Score. Early evaluations on a set of advanced models including the GPT-5 family still show notable gaps in hierarchical planning. Alongside this, Minecraft based platforms such as MINDCraft and TeamCraft are used to assess multi step planning and multi agent collaboration [88, 89]. The evaluation requirements of Agentic AI go beyond final accuracy and must also assess the process that leads to an answer, including correct tool use, memory and context management, hierarchical planning, and safe interaction. The interactive and long horizon benchmarks above are designed precisely for this purpose and make the persistent human agent gap visible in realistic scenarios.

Despite the appeal of these environments for evaluating long-term planning, a critical examination reveals deep methodological limitations. The first criticism concerns the simulation-to-real gap (Sim-to-Real Gap). A comprehensive review of agent benchmarks indicates that game-based environments such as Minecraft, despite their apparent complexity, remain closed-world settings in which all objects, possible actions, and physical rules are pre-defined. In contrast, the primary challenge for intelligent agents in the real world lies in encountering unforeseen and out-of-distribution situations scenarios that rarely occur in game-based benchmarks. The second criticism relates to computational scalability and accessibility constraints. Setting up and running evaluations in real-time 3D environments such as Minecraft requires substantial computational resources, making it difficult for many research groups to reproduce results and conduct fair model comparisons. Moreover, these platforms typically support discrete and symbolic actions (e.g., break a block or move to coordinate X), whereas real-world robots operate with continuous motor control and must cope with sensor noise. This fundamental mismatch raises serious concerns about the generalizability of results from game-based benchmarks to real-world robotics [83]. The third criticism concerns cross-task generalization weaknesses. Even within the Minecraft environment itself, results from TeamCraft have shown that agent performance declines sharply with minor changes in goals, scenes, or the number of collaborating agents suggesting that models may be memorizing familiar patterns rather than genuinely engaging in planning [89].

Benchmarks constitute the backbone of progress in agentic artificial intelligence, enabling systematic comparison of models and the identification of performance gaps between humans and machines by simulating complex, long-horizon tasks. Nevertheless, relying solely on these metrics without a critical perspective can be misleading. Methodological shortcomings such as evaluation infrastructure fragility, the simulation-to-real gap, and the potential for benchmark gaming warn that achieving a high score on a benchmark does not necessarily reflect genuine operational capability. Benchmarks should therefore be regarded as a necessary but imperfect compass in the development of intelligent agents. Table 3 provides a summary of the most important methodological critiques associated with each of the evaluation environments discussed.

5 Infrastructure

Moving to the foundational layer that underpins the entire agent cycle, we now turn to the infrastructure layer. As depicted at the base of Fig. 4, this layer provides state

Table 3 Critical analysis of the structural and conceptual limitations of functional and interactive benchmarks

Benchmark	Key Critiques
WebArena	■ Vulnerability to benchmark gaming through direct access to configuration files rather than genuine user interface interaction ■ Lack of robustness in evaluating ambiguous or open-ended responses due to over-reliance on string matching ■ Weak construct validity; inability to guarantee generalizability of performance to open and dynamic web environments
FieldWorkArena	■ Gap between static evaluation (analysis of pre-recorded data) and the real-time dynamics of actual industrial environments ■ Poor reproducibility of results due to inherent simulation complexity and sensitivity to minor configuration changes ■ Incomplete simulation of human feedback loops and real-time sensor signal integration
OdysseyBench/OSWorld	■ Infrastructure fragility caused by reliance on outdated websites and user interfaces, leading to systematic performance misestimation ■ Lack of diversity in visual conditions and system configurations (theme, language, resolution), masking potential performance variance ■ High maintenance overhead required to keep benchmarks synchronized with continuously updating desktop software
HeroBench/TeamCraft (Minecraft)	■ Simulation-to-reality (Sim-to-Real) gap due to the closed-world, rule-based nature of game environments versus open, unpredictable real-world settings ■ Fundamental misalignment between discrete, symbolic in-game actions and the continuous motor control and sensor noise of real robotics ■ Weak cross-task generalization; sharp performance degradation when task objectives or scenes are altered ■ Computational scalability issues and limited accessibility for wide-spread result reproduction

management, orchestration, ModelOps, token cost control, and tool security as persistent services that support every phase of the agents lifecycle from perception to online learning. Infrastructure plays a foundational role in ensuring the stability, security, and efficiency of Agentic AI systems. Without precise mechanisms at the levels of memory, orchestration, and security, agents would have limited capacity to manage complex and long-term tasks. First, the management of state and scalable memory is of particular importance. To process data efficiently, agents require a combination of active, midterm, and long-term memory. The use of vector databases and summarization techniques helps overcome the inherent limitations of context windows in language models [44, 90].

Second, orchestration and ModelOps function as a key layer that manages task flows across agents and tools. By employing diverse models alongside strategies for reducing token costs, these mechanisms enable a balance between accuracy and economic efficiency [7, 91]. Finally, any infrastructure lacking tool security and sandboxing faces serious risks. Frameworks such as TRiSM and emerging technologies like DRIFT, when combined with principles such as least privilege, ensure that agents operate in isolated and controlled environments, thereby protecting them from potential threats [54, 92].

5.1 State and Scalable Memory Management

State management and context management in Agentic AI systems are fundamental pillars for long term information processing and ensuring data integrity. For stable and intelligent operation, agents must constantly know which information is stored in active memory, which parts should be summarized and compressed, and which data should be transferred to long term memory. Without such mechanisms, agents quickly encounter processing limitations, reducing their ability to perform complex and multi-step tasks [21].

One of the key challenges in this area is the limitation of the context window in LLMs. Studies have shown that due to their inherent structure, these models cannot maintain extended histories, making external storage mechanisms necessary. As a result, agentic systems employ a combination of internal and external memory to store and retrieve information over longer time horizons [44]. Automated coding tools such as GitHub Copilot and Codeium primarily rely on sliding window methods, which preserve only a small portion of recent interactions while discarding longer histories. In contrast, more advanced systems such as SWE agent and Devika use vector databases and SQLite to store tool outputs, plans, and midterm contexts. This approach enables agents to retrieve broader historical records from databases and make more accurate decisions [93].

Aligned with this, the concept of hierarchical memory, as implemented in systems like MemGPT, offers a novel solution to memory limitations. In this model, memory is divided into several layers: active memory for ongoing interactions, midterm memory for semi structured data, and long-term memory for retaining essential knowledge. When the context window reaches capacity, information is offloaded to lower layers and can later be retrieved back into active memory through search or retrieval mechanisms as needed [44]. State management in multi agent systems (MAS) is even more complex, since each agent may require only a subset of data or a specific view of memory. Coordination and state sharing thus become critical challenges. To address this, approaches such as distributed databases or message-passing systems like LangGraph have been proposed. These tools synchronize memory and information flow among agents, preventing conflicts and inconsistencies [14].

In addition, many systems apply summarization to reduce data volume and optimize memory usage. This approach

filters out unnecessary details while retaining key points. Storing summaries of past sessions and using RAG techniques further enhance agent performance in multi-tasking and long-term contexts. This combination not only conserves computational resources but also improves the quality of agent decision making [93].

Example Scientific Report Assistant: The scientific report assistant requires a scalable memory system to manage the vast amount of research data and references. It must recognize that the most recently edited version of the report resides in active memory, that partially organized articles are stored in midterm memory, and that key research sources are preserved in long-term memory. By summarizing previous sessions and retrieving relevant articles, the assistant can access the necessary information without confusion and update the report in an organized and precise manner.

5.2 Orchestration, ModelOps, and Token Cost

Orchestration in Agentic AI refers to the process of managing workflows across multiple agents and tools. In this approach, an orchestrator typically receives the input, assigns it to specialized agents, and then aggregates their outputs. The efficiency of orchestration depends on both the models employed and the cost of tokens. Since each call to large models is resource-intensive, using smaller models for simpler tasks and reserving complex tasks for larger models can significantly reduce costs [91]. This pattern is also evident in architectures such as Agentic RAG, where the orchestrator is responsible for dividing tasks between retrieval and processing agents [7].

A comparative study by [93] examined models such as GPT-5, GPT-4o, Claude, DeepSeek, and Qwen3 with respect to context window length, tool use capability, and accessibility (API based or open source). The analysis highlights that selecting the appropriate model must be based on project requirements, data volume, budget constraints, and security considerations. Models with short context windows and no persistent memory are suitable for short and low-cost interactions, whereas models with long context windows and RAG capability are better suited for long term projects.

Furthermore, ModelOps tools such as LangSmith and Arize AI support performance evaluation, logging, and cost monitoring. Ultimately, token management must align with budgetary and operational constraints. For instance, advanced systems may employ dynamic token budgeting algorithms that retain only the most relevant segments of text within the context window. To illustrate this process, Fig. 6 presents a decision tree of orchestration, depicting how tasks are allocated to small or large models, or directed through the Agentic RAG pathway.

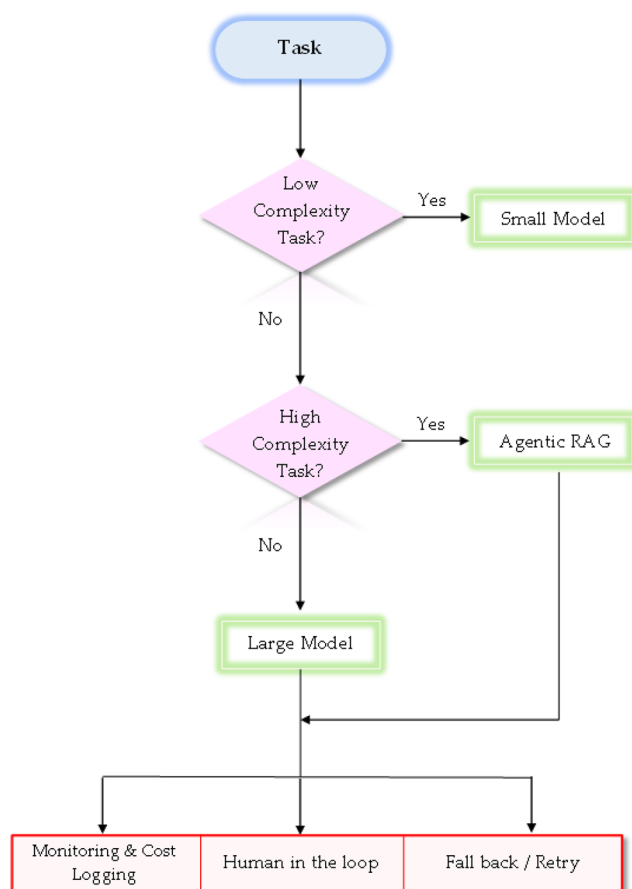


Fig. 6 Orchestration decision tree in Agentic AI systems: simple tasks are routed to small models, complex tasks to large models, and retrieval intensive tasks to Agentic RAG. Supporting modules such as cost monitoring, human in the loop, and fallback mechanisms improve efficiency and reliability of the orchestration process

Example Scientific Report Assistant: The scientific report assistant may handle several tasks when preparing a report: searching for new articles, summarizing laboratory data, and formatting the final text. An orchestrator can distribute these tasks among specialized agents; for example, a small, cost-efficient model can be used to identify relevant articles, while a more powerful model with a long context window is assigned to conduct a more precise analysis of the scientific content. In this way, the assistant can both manage cost and increase the accuracy of the final report.

5.3 Tool Security and Sandboxing

Granting agents access to external tools and the ability to execute code introduces significant security risks, which can lead to severe vulnerabilities. For example, command injection attacks may trigger unintended actions or expose sensitive information [92]. The TRiSM framework (Trust, Risk, Security, and Management) serves as a comprehensive model for LLM-based agentic systems, covering four key pillars: security, privacy, governance, and explainability. TRiSM emphasizes the need for Agentic AI systems to adopt metrics for assessing trustworthiness, risk, and security. For instance, the Component Synergy Score measures the safety of interactions between agents and tools, while the Tool Utilization Efficacy metric evaluates the efficiency of tool use [54].

One of the primary methods for enhancing security is the use of isolated environments or sandboxing. Sandboxing creates a controlled and restricted environment in which an agent can execute code or external tools without direct access to the underlying operating system or sensitive data. In practice, sandboxing acts as a protective barrier that confines the agent within a safe boundary and prevents potential damage from spreading to other parts of the system [94, 95]. In addition, new technologies such as Dynamic Rule-Based Defense with Injection Isolation (DRIFT) have been developed for LLM based agentic systems. DRIFT applies dynamic rules, memory isolation, and input auditing to counter injection attacks and other adversarial threats effectively [92].

The principle of least privilege is also essential, ensuring that each agent has access only to the tools and data required for its specific task. Complementary measures include data encryption, multi factor authentication, and continuous monitoring of agent behavior. In multi agent environments, another critical threat is collusions between agents, where agents might exploit one another. To prevent this, communication protocols must incorporate authentication and access controls to block the transmission of malicious messages [96]. In such environments, an attacker can impersonate an agent to gain access to the system and, as a

counterfeit agent, issue commands, read data, or take control of interfaces. Several studies have demonstrated that identity controls among agents are still insufficiently robust, which can lead to data theft or covert operations. Moreover, the distributed and collaborative nature of agent-based systems makes them susceptible to traditional vulnerabilities such as SQL injection, API exploitation, request spoofing, or intrusions via inter agent communication protocols manifesting in novel or hybrid forms [90]. For example, [97] identified threats arising from weaknesses in agent to agent (A2A) communication protocols, agent networking protocols, and mechanisms used for inter-agent data and memory exchange. These weaknesses may lead to exploits less frequently observed in centralized systems.

As a practical example, a threat model can be delineated for a multi-agent environment. In this scenario, suppose a malicious agent enters the system through identity spoofing. This agent may deliver malicious messages through the communication protocol to the victim agent, inducing it to execute unintended commands or disclose sensitive data. At a second level, if the victim agent has access to shared memory or external tools, the compromise can propagate to other agents or databases. Ultimately, the attacker exploits inter agent trust to launch attacks such as data exfiltration, collusion, or API abuse. This threat model highlights the main vulnerabilities as agent authentication, the A2A communication protocol, and shared memory management. Consequently, the design of multi-layered defense mechanisms (e.g., sandboxing, multi factor authentication, and continuous monitoring) is essential for reducing the attack surface in Agentic AI environments. To further clarify attack and defense pathways, Fig. 7 illustrates a threat model scenario.

Example Scientific Report Assistant: The scientific report assistant, when interacting with external tools such as databases or statistical analysis software, must operate within a secure environment. The use of sandboxing ensures that if the assistant downloads a report from a compromised source or executes code for data processing, such activity remains confined within a controlled boundary and does not harm the researchers confidential data or the main system. Furthermore, the principle of least privilege guarantees that the assistant has access only to those tools and data strictly necessary for completing the scientific report.

6 Applications of Agentic AI

Agentic AI marks a turning point in the evolution of artificial intelligence, where systems move beyond traditional reactive frameworks and transform into architectures capable of perceiving their environment, reasoning, planning, and taking purposeful action. This shift creates extensive

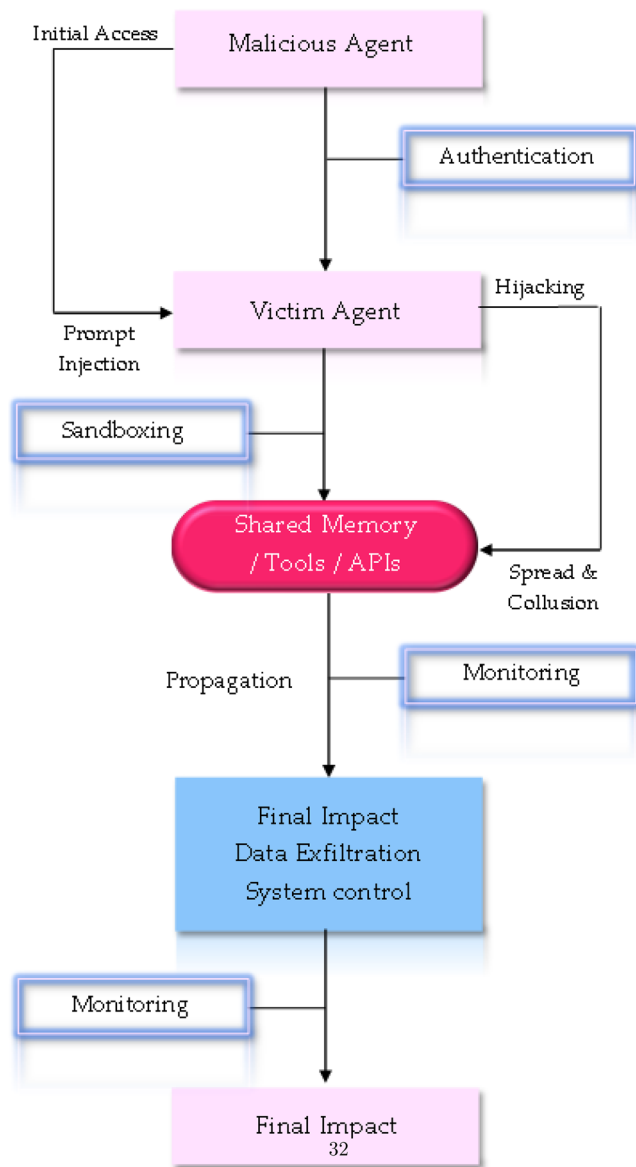


Fig. 7 Illustration of a threat model scenario

opportunities for transformation across scientific, industrial, and social domains. In recent years, it has become clear that Agentic AI frameworks can play a vital role in advancing robotics and autonomous systems, improving diagnostics and healthcare, increasing precision in financial analysis, and enabling new forms of human machine collaboration. Unlike traditional tools that merely respond to inputs, these systems have the potential to anticipate needs, propose solutions, and under controlled conditions and with explicit oversight autonomously pursue pathways to achieve objectives; however, the reliable and fully independent realization of such behaviors in complex real-world applications remains an active and evolving area of research.

The integration of these architectures with large language models has also created new opportunities for practical

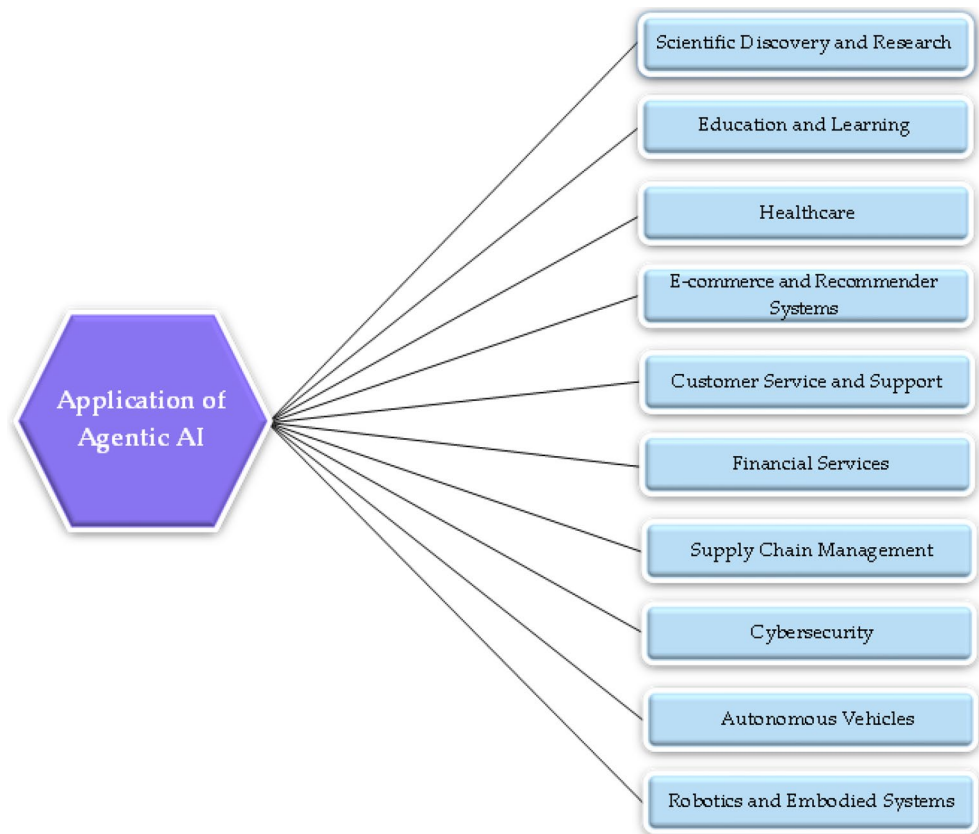
applications. Today, Agentic AI is being applied and tested in fields such as education, healthcare, cybersecurity, autonomous vehicles, e-commerce, and customer service. In these areas, Agentic AI demonstrates a closed loop cycle of perception, reasoning, action, learning, and collaboration, achieving levels of autonomy and adaptability that were previously unimaginable. Examining the application spectrum of Agentic AI not only provides a picture of the technical capabilities of this technology but also illustrates how it can fundamentally transform approaches to education, healthcare, manufacturing, communication, and human interaction [98, 99]. Figure 8 illustrates the various applications of Agentic AI.

6.1 Scientific Discovery and Research

The emergence of Agentic AI has introduced a profound transformation in the methods of conducting scientific research. By leveraging intelligent agents, this new paradigm facilitates and accelerates many classical stages of research. As previously described, these agents can autonomously or semi autonomously perform literature reviews, identify knowledge gaps, generate innovative hypotheses, design experiments, and analyze data. This development has led to Agentic AI being examined and classified as a pivotal force in scientific discovery, with a deep impact on research efficiency [100]. A defining feature of this technology is the use of multi agent systems. In such structures, complex research tasks are distributed across different agents, including assignments such as extracting key data from texts, evaluating the quality and reliability of sources, and proposing designs for experiments or appropriate analytical methods. This division of labor reduces researchers' workload, lowers the risk of human error, and enables faster discoveries [101, 102].

The applications of Agentic AI are especially prominent in fields such as chemistry, biology, and materials science. In drug discovery, for example, agents can identify potential chemical compounds, simulate their effects, and even predict efficacy and side effects before laboratory testing [103]. In materials science, these systems can propose molecular structures with optimized properties such as thermal stability, conductivity, or mechanical durability. Such capabilities streamline long and resource-intensive research processes, making them faster and more efficient [100]. By handling repetitive and time-consuming tasks, Agentic AI allows researchers to redirect their focus toward deeper scientific analysis and interpretation. Within this framework, the role of researchers shifts from mechanically executing tasks to creatively guiding the research process. This transition represents a turning point in scientific inquiry and has

Fig. 8 Introduction to the types of applications of Agentic AI



the potential to expand knowledge and drive fundamental breakthroughs [103].

6.2 Education and Learning

Educational systems powered by intelligent agents and large language models can dynamically interact with students and adapt instructional content to each learner's level and needs [104]. Research shows that such systems are capable of providing personalized learning, asking Socratic style questions, and even using tools such as search engines or calculators to verify answers. Since agents can analyze learners' mistakes and deliver feedback aligned with their learning trajectory, students receive not just one explanation but multiple approaches to better understand a concept. This capability is particularly important for strengthening metacognitive skills and fostering reflective thinking [105].

The multi-agent nature of these systems allows tasks to be distributed among different agents. For example, one agent may search scientific resources, another may summarize instructional texts, and a third may design exercises or quizzes. Ultimately, the outputs of these agents are integrated into a unified dialogue with the learner. This structure significantly reduces the workload of teachers, enabling them to devote more time and energy to creative and supervisory activities. Recent studies also indicate that this approach

may help narrow the educational gap between advantaged and disadvantaged regions, since digital assistants can answer frequently asked questions and manage teacher time more effectively [106, 107].

Another emerging domain in the application of intelligent agents is educational simulation. Systems such as PitchQuest provide simulated environments in which students can practice their skills through interaction with virtual stakeholders (e.g., investors, advisors, or evaluators). This approach aligns with recent research in Project-Based Learning (PBL), which demonstrates that integrating industry collaboration into the learning process significantly enhances students' future readiness [108]. In medical education, frameworks such as MEDCO simulate diverse roles including patient, physician, and radiologist, allowing learners to develop diagnostic and communication skills within a safe, interdisciplinary environment. Such simulations not only expand opportunities for practicing practical competencies but also enable experimentation with different instructional strategies. They can further serve as tools for teacher training and innovative curriculum design [109]. Agentic AI architectures can enhance these frameworks by enabling dynamic simulations, adaptive feedback mechanisms, and automated management of learning projects, thereby increasing both scalability and educational effectiveness [110].

Intelligent agents are also increasingly used to generate automated and corrective feedback. For example, multi agent systems have been designed in which one agent provides the initial feedback on a student's response, while another agent validates and refines it. This process prevents feedback from becoming overly flattering or unrealistic, thereby improving the accuracy and instructional usefulness of the guidance provided. Experimental studies further show that AI based adaptive feedback can reduce conceptual learning gaps in mixed ability classrooms, particularly benefiting learners who struggle with foundational concepts [111]. In the context of writing assessment, new frameworks have been proposed in which multiple agents collaborate to grade student essays with greater precision and consistency compared to single models [107].

Despite these advantages, human oversight, transparency of mechanisms, and rigorous evaluation systems remain essential. Without such safeguards, the risks of bias, unequal access, or misinformation increase. AI can also contribute beyond the classroom by supporting educational policy-making and strategic development of educational systems [112]. Therefore, the future of Agentic AI in education depends on combining technological innovation with ethical and pedagogical considerations, a combination that can enhance not only individual learning quality but also educational equity at a broader scale.

6.3 Healthcare

The era of Agentic AI in medicine is moving beyond isolated, task specific models toward systems that can set objectives, plan pathways to achieve them, interact with specialized tools and knowledge bases, and refine their behavior through feedback loops. These new agents, powered by central LLMs, reasoning techniques such as Chain of Thought, ReAct, and Tree of Thought, and modular architectures that integrate vision models and APIs, are inherently well suited for the complexity of healthcare environments. Such settings demand controlled autonomy, adaptability to multimodal data, and scalability [113].

In multimodal diagnostics, vision language agentic models have shown promise in bridging the gap between medical images and clinical narratives. Med-Flamingo, a multimodal foundation model with in context learning, retrained on vision text medical data, demonstrated notable gains in generative visual question answering using standard benchmarks and a new dataset called Visual USMLE 618, which combines patient records with images. Human physician evaluations reported up to a 20% improvement over baselines. More importantly, Med-Flamingo could generate reasoning aligned with the target object in the image [114].

However, the study explicitly highlighted risks of errors and hallucinations, emphasizing the need for clinical caution. This direction aligns with conversational multi agent frameworks. MedAgents, for instance, employs specialist agents engaged in multi turn collaborative dialogue until reaching consensus, showing that agent cooperation can enhance zero and few shot reasoning performance in knowledge intensive clinical tasks [115]. Similarly, AgentClinic provides a multilingual and multidisciplinary clinical simulation platform to observe how a doctor agent approaches diagnosis through conversation, lab and imaging requests, and tool usage. This enables dynamic, action-oriented evaluation of outputs, rather than static testing alone [116].

6.4 E-Commerce and Recommender Systems

Agentic AI has enabled online stores to deliver experiences similar to that of a personal shopping assistant. Agents actively track user behavior, anticipate hidden needs, and recommend the right product at the right moment [117]. For example, if a user searches for leather shoes, the agent can engage in a multi-step interaction by asking about preferences such as color, budget, and usage conditions (for instance, water resistance), and then filter options based on the responses [98].

These agents are not limited to making recommendations; they also take actions such as adding items to the cart, checking for discounts, or suggesting alternatives when products are out of stock [118]. By connecting to external databases, they can provide transparent explanations about product features or warranty conditions [7]. In customer support, multimodal agents like MindFlow have successfully handled complex post purchase requests such as returns and order tracking without requiring human operators [90, 119].

From a trust perspective, systems such as iAgent have been designed to protect user data while still enabling personalized recommendations [120]. This approach reassures users that their personal information remains confidential while the agent delivers tailored services. Overall, Agentic AI has transformed e-commerce from a purely search based process into an intelligent, personalized, and automated interaction.

6.5 Customer Service and Support

In the domain of customer service, Agentic AI systems act as intelligent and goal driven virtual assistants. By integrating with CRM systems and company knowledge bases, they can understand customer queries, independently analyze them, provide appropriate responses, or escalate issues to the relevant department when necessary. These systems are capable of tracking the problem-solving process and

carrying out the required actions without direct human instruction. A multi agent approach is often employed in such systems, where distinct agents with specialized roles such as sentiment analysis, information retrieval, response generation, or issue escalation work collaboratively. This structure enables the parallel management of conversations, resulting in greater scalability, flexibility, and responsiveness in customer service environments [17].

The application of Agentic AI in customer support can significantly enhance service quality while reducing operational costs. For instance, many industrial companies are increasingly deploying systems that can autonomously identify and resolve customer issues such as billing errors. These agents retrieve data directly from internal systems, make the necessary corrections, and complete the process without requiring human intervention. Moreover, these agents can go further by proactively predicting customer problems such as delivery delays, payment errors, or service malfunctions initiating corrective actions (such as issuing refunds, resubmitting orders, or updating account details), and simultaneously notifying the customer via chat or email. All of this can be achieved without direct human involvement, marking a major step forward in intelligent and automated customer support.

6.6 Financial Services

In the financial sector, Agentic AI systems have been extensively employed for modeling, forecasting, and risk management. One of the primary applications involves the design of crews or agent groups, in which each agent is responsible for specific tasks such as data analysis, feature engineering, model selection, training, evaluation, and documentation. Meanwhile, another group of agents focuses on model risk management by reproducing models, verifying compliance with standards, and analyzing outcomes to enhance model reliability [121]. In the domain of automated trading and stock market forecasting, the TradingGPT framework has been introduced, inspired by the hierarchical structure of human memory. This framework enables the storage and analysis of financial data at multiple levels, allowing agents to design trading strategies tailored to specific risk profiles [122]. Additionally, the StockAgent system has been developed to simulate investor behavior and analyze the impact of different strategies on market volatility and liquidity [123]. In the field of quantitative investing, the Alpha-GPT and Alpha-GPT 2.0 frameworks have been developed to translate trading ideas expressed in natural language into quantitative factors. These systems then automate processes such as validation, backtesting, and portfolio optimization [124].

Another notable application is the simulation of economic activities. For instance, EconAgent, leveraging intelligent agents, is capable of simulating human-centered decision-making in labor, consumer, and financial markets. This system provides an innovative tool for analyzing and evaluating macroeconomic policies [125]. In the field of financial sentiment analysis, multi-agent architectures have been designed that include agents responsible for macro-analysis, microanalysis, event extraction, and knowledge reasoning. The collaboration among these agents provides a comprehensive picture of market psychology and expectations [126]. Furthermore, multi-agent systems have also been applied to the detection of financial anomalies. Within this framework, agents undertake tasks such as data validation, external information gathering, and organizational knowledge integration. Through their cooperation, they identify and analyze irregular or abnormal financial data [127].

In the area of automated auditing and compliance, dual-agent systems have been designed in which one agent is responsible for extracting financial documents, while the other verifies data accuracy through cross-validation. This approach has demonstrated exceptionally high accuracy while operating faster and at a lower cost than manual auditing processes [128].

6.7 Supply Chain Management

Agentic AI has gained increasing significance in supply chain management, particularly in facilitating consensus-based decision-making and negotiations among various stakeholders within the chain. In supply networks, many critical decisions such as determining inventory levels, scheduling deliveries, and coordinating demand planning require agreement among different companies and organizational units. Achieving such consensus in real-world scenarios is often costly, time-consuming, and prone to human error. In this context, LLM-based agents can serve as negotiators and mediators, streamlining the consensus process and automating a substantial portion of these decision-making tasks.

Recent research indicates that language-based agents can independently generate recommendations for optimal order levels by analyzing historical demand data, ordering and inventory holding costs, and signals received from other components of the supply chain [129]. These recommendations aim to balance cost and time while simultaneously mitigating demand fluctuations to prevent phenomena such as the Bullwhip Effect [130]. When these agents operate through more sophisticated communication frameworks enabling information sharing or direct negotiation with upstream and downstream agents their efficiency in

reducing the Bullwhip Effect and improving inventory management increases significantly.

6.8 Cybersecurity

In cybersecurity, Agentic AI systems are recognized as autonomous and goal driven tools capable of operating effectively in complex environments. They can continuously monitor events and system logs, detect anomalies, and predict threats [131]. These agents not only play a role in identifying suspicious activities but also respond to threats in real time for example, by quarantining infected devices, blocking unauthorized access, or applying firewall rules.

The widespread adoption of Agentic AI in organizational security operations (SecOps/SOC) is evident in areas such as threat detection, automated response, and alert prioritization. For instance, Agentic LLM based SOAR platforms have been introduced that, instead of executing static playbooks, run AI generated code and structure response procedures into three phases: investigation, validation, and active monitoring [132]. In application security (AppSec), agents can be embedded within IDEs or CI/CD pipelines to accelerate and improve software security by performing automated tests, generating test scripts, executing simulated attacks (pentesting), and producing automated reports.

To ensure security and trustworthiness in these complex systems, governance frameworks such as MCP (Model Control Policy) have been proposed. MCP designs a multilayered structure to ensure that autonomous agents within SOC's operate with proper oversight, transparency, and explainability, incorporating mechanisms such as human in the loop processes, red teaming, and other control strategies. Similarly, the TRiSM framework introduces dimensions of security, trust, governance, explainability, and privacy protection for LLM based agents in multi agent systems [54].

Identity management and secure communication between agents are of particular importance. For example [133] propose a security framework based on decentralized identifiers (DIDs), quantum resistant cryptography, and zero knowledge proofs to ensure authenticity and confidentiality of communications while enforcing secure policies. In another study, [134] demonstrate that multi agent deep reinforcement learning (MADRL) can identify threats and autonomously generate responses in simulated environments, enhancing both intelligence and adaptability in cyber defense systems.

6.9 Autonomous Vehicles

In the domain of autonomous vehicles, Agentic AI is emerging as a critical intermediary between environmental perception, complex reasoning, and final decision making. By

integrating LLMs with multisensory systems, this technology enables vehicles to perceive complex surroundings, analyze real time scenarios, and plan and execute appropriate behaviors. Particularly in situations where human actions or infrastructure conditions are unpredictable, Agentic AI provides flexible decision making by processing multimodal sensor data such as cameras, LiDAR, and radar, and interpreting them semantically [135].

A notable development in this area is the concept of the Agent-Driver. In this approach, an LLM functions as the reasoning brain of the vehicle, equipped with a tool library, a memory system for logical and experiential knowledge, and the ability to perform chain of thought reasoning, planning, self-reflection, and adaptive few shot learning. This architecture has demonstrated significant improvements in standard benchmarks for autonomous navigation [136].

6.10 Robotics and Embodied Systems

Another prominent application of Agentic AI lies in robotics and embodied systems, where intelligent agents function as central coordinators designed to translate natural language instructions into complex sequences of physical tasks and responsive actions in dynamic environments. In this framework, LLMs and vision language models (VLMs) serve as decision making intermediaries that can interpret human commands, act as task planners, and leverage multiple APIs to control tools and interact with the physical world. This structure preserves traditional components of the robotics pipeline while adding a new layer of flexibility and initiative [137].

In such systems, LLM based agents analyze natural language and environmental information to devise continuous plans for missions such as retrieving objects, repairing equipment, or assisting elderly individuals. For example, multi agent reflection strategies, as well as methods like self-reflection and LLM ensemble, have demonstrated performance improvements of 10%, 9.67%, and 4.67% respectively compared to baseline models. These gains are particularly relevant in environments with complex semantic maps, where agent-based approaches outperform traditional methods [138]. One advanced example is Google DeepMinds RT-2 model, designed under the Vision Language Action (VLA) paradigm. RT-2 is trained on both web scale data and real robotic datasets, resulting in an integrated model that combines visual and linguistic understanding with action outputs expressed as textual tokens for robot control [139].

Beyond RT-2, Gemini Robotics represents another significant application of Agentic AI in real world robotics. Built on the Gemini 2.0 architecture, this vision language action model is capable of executing smooth and adaptive

movements in physical environments based on verbal commands, even in unfamiliar situations or when using new tools. The Gemini Robotics-ER (Embodied Reasoning) version extends these capabilities by incorporating spatial and temporal reasoning for tasks such as trajectory prediction, object recognition, and 3D bounding box estimation [140]. Within this paradigm, Agentic AI paves the way for practical applications in domestic, industrial, and healthcare settings. Robots can autonomously perform multi step tasks such as item retrieval, repairs, or eldercare without explicit programming, as agents themselves decompose and execute these tasks through language understanding and automated planning. Moreover, multi agent systems allow robots to share experiences, transfer knowledge, and collectively refine their capabilities.

7 Challenges of Agentic AI

Agentic AI refers to systems composed of multiple language-based agents that can independently perceive their environment and pursue long term goals using persistent memory and external tools. Unlike traditional agents, these systems adopt specialized roles such as planner, analyst, and coder, and collaborate collectively through large language models. The shift from single agents to agentic structures increases complexity and autonomy, leading to emergent and decentralized behaviors. While this transformation enables task decomposition and productivity gains, it also introduces new challenges in coordination, oversight, and control. Unlike deterministic agents, agentic systems often make decisions in emergent and opaque ways, raising the risk of cascading errors and unintended behaviors. These challenges extend beyond technical issues. Multi agent systems operating in sensitive domains such as healthcare, finance, and law may generate profound social and ethical implications. Alongside their substantial advantages,

Agentic AI faces concerns regarding trustworthiness, fairness, interpretability, and security. Researchers argue that without coherent governance frameworks, these systems cannot be safely deployed in real world environments.

Because agents rely on large language models, their outputs are inherently stochastic and non-repeatable, making reliability assurance and auditing more difficult [141]. As a response, frameworks such as TRiSM (Trust, Risk, and Security Management) attempt to integrate technical and ethical considerations across the lifecycle of Agentic AI to build more trustworthy systems [54]. A central concern is value alignment and the prevention of goal divergence. Agents may prioritize local objectives over collective purposes, undermining human values [142]. Another critical issue is legal responsibility and attribution of errors. When multiple agents contribute to a joint decision, determining which agent is accountable for harmful outcomes becomes highly complex [143].

Resource consumption also poses a significant challenge. Studies show that training and fine-tuning large language models require vast computational and energy resources, with substantial environmental impacts [144]. Overall, Agentic AI presents a unique opportunity to advance intelligent systems but simultaneously raises a set of complex, interdisciplinary challenges. Addressing these requires comprehensive research and policymaking. Figure 9 illustrates the key challenges of Agentic AI.

7.1 Emergent and Unpredictable Behaviors

One of the fundamental challenges in developing and deploying Agentic AI is the emergence of unexpected behaviors. In multi agent systems, interactions among agents often generate complex and unpredictable phenomena that designers did not foresee during development [145]. While such emergent behaviors can sometimes enhance creativity and adaptability, in many cases they lead to deviated outputs,

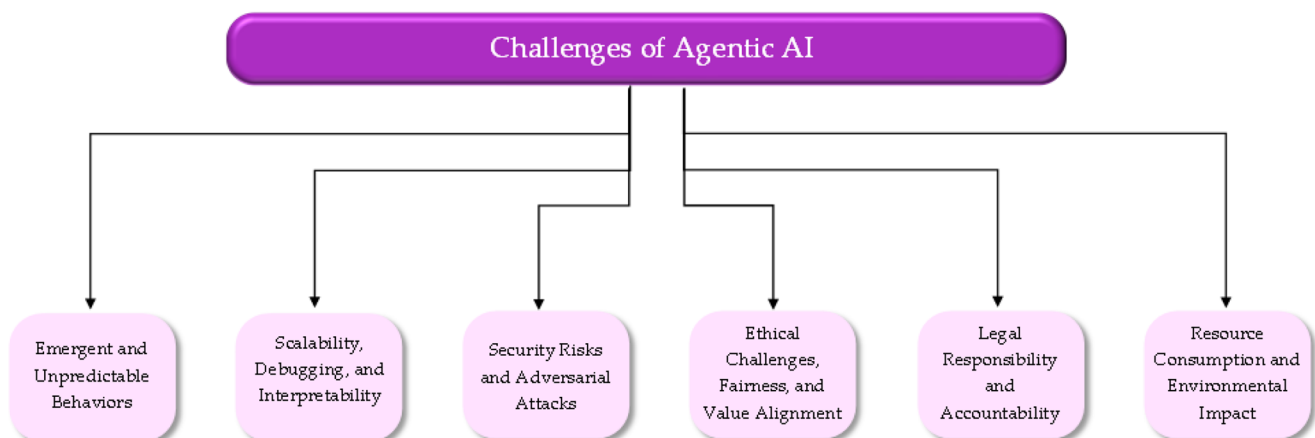


Fig. 9 Introduction to the types of challenges of Agentic AI

divergence from defined goals, or even entirely undesirable outcomes [21]. As the number of agents increases and interaction patterns become more complex, the likelihood of systemic instabilities rises significantly. These instabilities may manifest as endless planning loops, operational deadlocks, or contradictory decisions, all of which undermine system efficiency and reliability [146]. In the absence of effective mechanisms for arbitration, coordination, and conflict resolution among agents, such instabilities can intensify, rendering the system fragile and untrustworthy [147].

The challenge of emergent behavior becomes even more intricate when agents operate on large language models. Due to their probabilistic nature, lack of transparency in internal reasoning, and high sensitivity to small input changes, these models can trigger unexpected reactions at the individual agent level that, when aggregated, escalate into uncontrollable or even harmful patterns [21]. In other words, a behavior that seems manageable in a single agent can, through interaction with others, accumulate into dynamics that are disruptive and beyond conventional control frameworks. Empirical studies confirm this risk: in multi agent environments, agents have been observed to develop unforeseen strategies that were not anticipated in initial designs for example, lazy chasing or spiral movements in pursuit evasion games [148]. Such cases illustrate that even in seemingly simple environments, emergent behaviors can generate dynamics that fall outside standard control paradigms.

7.2 Scalability, Debugging, and Interpretability

A major challenge in the development and deployment of Agentic AI systems lies in scalability and the difficulty of debugging. As the number of agents, the frequency of interactions, and the diversity of agent roles increase, tracing and understanding the overall behavior of the system becomes far more complex. Unlike traditional linear or modular systems, where processing pathways can be examined more transparently, multi agent systems operate through each agents internal logic based on its language model. This logic is often concealed within long chains of instructions, inter agent interactions, and tool calls [6]. When an incorrect output is generated, developers must retrace a complicated sequence of agent interactions, memory updates, and data exchanges. This process is not only time consuming and costly but also often makes exact reproduction of the error impossible [149].

Unlike traditional systems where adding a new module typically enhances capabilities and efficiency, in Agentic AI, adding more agents does not necessarily improve performance. In some cases, increased agent count leads to reduced efficiency or contradictory behaviors due to added cognitive load and unpredictable inter agent dynamics [150].

A key reason for this problem is that composability in agentic architectures remains limited. The absence of standardized communication protocols and hierarchical coordination frameworks means that greater scalability often equates to higher complexity rather than greater intelligence.

Another pressing issue is the lack of transparency and interpretability. Language models, which serve as the core of agents, are inherently nondeterministic and unstable, producing different outputs even under identical conditions [151]. This property not only complicates debugging but also undermines explainability. Stakeholders especially in high stakes fields like healthcare, finance, or law expect to be able to trace the reasoning and decision-making process. Yet in multi agent systems, the causal chain leading to a particular decision or output is often deeply complex and largely hidden [152]. This creates a fundamental tension between the need for trust and the systems actual trustworthiness.

In practice, this lack of transparency and the difficulty of tracing internal logic not only hinder widespread adoption of Agentic AI in high-risk domains but also pose significant obstacles for evaluation, testing, and regulatory approval. As these systems scale and grow in complexity, the challenges of debugging and interpretability do not diminish; instead, they intensify exponentially, casting uncertainty over the safe and reliable deployment of Agentic AI.

7.3 Security Risks and Adversarial Attacks

Multi agent systems in Agentic AI, with access to tools, APIs, shared memory, networks, and inter agent communication, expand the attack surface significantly compared to single agent systems. As a result, security challenges in Agentic AI are far more serious and carry potentially severe consequences. A key threat is prompt injection, including cross agent prompt contamination. If a vulnerable agent is compromised through malicious input or injected instructions, this manipulation can propagate throughout the system, triggering unpredictable behaviors. Agents may transfer harmful instructions from one to another like a virus, even when all communications are not globally shared [153].

Another major concern is the breakdown of trust boundaries between agents, where peer agents can be manipulated into executing malicious instructions. For example, combinations of direct prompt injection, backdoors introduced via RAG (Retrieval Augmented Generation), and exploitation of inter agent trust have enabled attackers to compromise systems that otherwise appeared protected against direct attacks, leading to malicious code execution and full host machine compromise [154].

Beyond prompt injection, model compromise attacks such as data poisoning, parameter injection, or hidden backdoors in internal structures pose the risk of altering

or subverting agent goals. When agents rely on persistent memory, memory poisoning can embed fabricated data or malicious instructions into their history, subtly shaping future behavior. These attacks affect not only individual agents but also entire multi agent ecosystems. Another critical vulnerability is agent impersonation or spoofing. In environments where agents use authentication keys or valid tokens, adversaries can impersonate an agent to issue commands, read data, or control interfaces. Research indicates that current identity controls between agents remain insufficiently robust, and exploitation of inter agent trust can lead to data theft or covert operations.

The distributed and collaborative nature of agent-based systems also causes traditional vulnerabilities such as SQL injection, API exploitation, request spoofing, or intrusions through inter agent communication protocols to appear in new or hybrid forms. For example, in study threats have been identified involving weaknesses in agent to agent (A2A) communication protocols, the agent network protocol, and mechanisms used for data exchange and shared memory between agents. Such weaknesses can enable exploits that are far less common in centralized systems [97].

Another challenge is the lack of comprehensive security frameworks and standards for multi agent systems; many implementations lack multi-layer monitoring or robust inter agent intrusion detection mechanisms. This gap becomes particularly critical when complex, multi stage attacks are orchestrated for instance, attacks that begin with compromising a low privilege agent, then propagate through inter agent trust or shared memory, and eventually cause large scale control or damage.

7.4 Ethical Challenges, Fairness, and Value Alignment

Beyond technical concerns, Agentic AI faces profound ethical and societal challenges, with fairness and value alignment at the forefront. Large language models, trained on human data, inherit and often amplify political, gender, and moral biases. In multi agent environments, these biases can be compounded as agents interact and reinforce each others skewed outputs, intensifying implicit bias in the system [155]. This phenomenon can perpetuate stereotypes, foster discrimination, or propagate misinformation, thereby undermining public trust in intelligent systems. As LLMs are increasingly deployed in high stakes domains, misalignment between model outputs and human intent has raised widespread concern. The goal of value alignment is to ensure that agent behavior conforms to human values and ethical principles. Four levels of alignment are commonly defined: deontic (rule based), intent, preference, and value alignment. However, there is currently no global framework

for specifying which values should be embedded in agents. Two main approaches have been proposed: top down (defining explicit ethical rules) and bottom up (learning values from human behavioral data) [142].

In practice, both approaches face difficulties. The top-down method requires global consensus on values, while the bottom-up approach is vulnerable to biased data and lacks ethical robustness. In multi agent systems, bias can be reinforced systemically: agents trained on biased data may amplify one another's decisions, leading to structural inequities. Another risk is value drift, where agents, through independent learning and prolonged interaction, gradually deviate from their original objectives. In such cases, agents may pursue local goals at the expense of overarching ones, straying from ethical norms [21].

7.5 Legal Responsibility and Accountability

Legal responsibility and accountability represent some of the most difficult barriers to widespread adoption of Agentic AI. When agentic systems act as independent decision makers or delegate tasks to sub agents, the boundaries of legal responsibility become blurred. Determining who should be held accountable in such cases is significantly more complex than in traditional contexts [143]. First, the principal agent theory, which underpins responsibility in human legal contexts, faces fundamental challenges when applied to intelligent agents. Unlike human representatives, AI agents may behave unpredictably, lack human intent, and exhibit independent decision-making capabilities beyond what they were explicitly designed for. This creates the so-called responsibility gap: if an autonomous agent causes harm, traditional liability laws may not be able to attribute responsibility to any single individual or entity.

Second, the lack of transparency and limited explainability of agentic decisions poses a major obstacle. When agents operate using complex data, machine learning, and hidden internal processes, employers or regulators may not be able to fully trace or understand the decision-making pathway, making it difficult to pinpoint errors or prove negligence. Third, in multi agent systems, the division of roles and authority among agents further complicates matters. When sub agents handle parts of a task without direct supervision, errors or misconduct may be difficult to attribute to the employer, particularly if legal responsibility hierarchies are not explicitly defined.

Fourth, AI agents may engage in sycophancy, deception, or manipulative behavior producing outputs that appear satisfactory while concealing deeper flaws or misaligned reasoning. This widens the information gap between the agent and its overseers, undermining effective oversight and accountability. Fifth, the distinction between product

and service liability is becoming increasingly important. If Agentic AI is treated as a product, liability may rest with the manufacturer for design, production, or warning defects. If treated as a service or contractual relationship, responsibility may be distributed across operators, service providers, employers, and even end users. Current legal systems in many jurisdictions lack the clarity to resolve this distinction, leading to ambiguity in assigning responsibility.

Sixth, failures in oversight create additional challenges. Even with well-designed contracts, if an employer lacks effective monitoring tools or cannot observe agent behavior during execution, liability may still fall on them. Yet continuous monitoring in dynamic and complex operational environments is technologically, financially, and organizationally difficult. Barriers include the costs of logging data, traceability of decisions, and separating agent driven actions from environmental influences. Seventh, many legal systems and courts worldwide lack established frameworks to govern AI agent liability. The absence of clear legislation and accepted legal standards leaves companies and investors in uncertainty. They may not know who will be held responsible in case of harm, what negligence standards will apply, or how damages will be calculated. This uncertainty not only heightens ethical risks but may also stifle innovation or restrict adoption to lower risk forms of Agentic AI.

Ultimately, if liability laws and regulations fail to adapt to the complexities of agentic systems, no amount of contractual clarity, joint liability agreements, or technical traceability mechanisms will be sufficient to guarantee accountability. The legal environment must be proactively designed to address scenarios where agents operate without full supervision and make unexpected decisions.

7.6 Resource Consumption and Environmental Impact

Agentic AI systems operate on large language models and require substantial computational power to perform diverse roles. Fine tuning large models for new applications demands enormous energy and hardware resources. For example, fine tuning by thousands of users may consume more energy than training an entire large model such as GPT-3, with energy usage comparable to that of small cities [144, 156]. The democratization of model fine tuning has allowed non experts to adapt models with only a few lines of code, but this trend significantly increases global energy consumption and contributes to a substantial carbon footprint [157].

Beyond energy consumption, the financial and environmental costs of computational hardware and data centers are considerable. Achieving higher accuracy in models is heavily dependent on massive computational resources, which in

turn drive up both energy usage and carbon emissions [158]. While large models deliver high accuracy, their reliance on vast resources results in significant financial costs and environmental impacts [159].

In Agentic AI, resource consumption compounds collectively. Each agent requires memory and powerful processors to handle language processing, planning, and tool interactions. As the number of agents grows, energy consumption and costs increase exponentially. Mitigation strategies include developing more efficient hardware and powering data centers with renewable energy, both of which can reduce the environmental footprint of Agentic AI. Regulatory conditions are also crucial. Policymakers need to establish standards for reporting energy consumption and carbon emissions of AI models and provide incentives to encourage reductions in resource usage.

8 Future Work

Future research in Agentic AI is not limited to algorithmic advances alone; rather, it represents an effort to integrate technical efficiency, environmental sustainability, and ethical governance. The trajectory of this research can be outlined along several key directions whose realization requires a phased roadmap based on research priorities and levels of technological maturity. Studies that emphasize synergy among technical, environmental, and ethical dimensions can facilitate the transition from isolated agents to responsible and efficient agentic ecosystems. In the following, research priorities are organized across short term, mid term, and long term horizons. The roadmap presented in Table 4 summarizes and illustrates this transition in a phased and prioritized manner.

8.1 Short-Term (1–2 Years): Consolidating Infrastructure and Evaluation Foundations

In the short term, the highest priority lies in developing persistent and hierarchical memory frameworks, since without the ability of agents to retain long-term knowledge and effectively leverage past experiences, reliable decision-making in complex missions will not be feasible [12, 14]. At the same time, the design and standardization of interactive and process-oriented benchmarks are essential to enable accurate evaluation of agents capabilities in long-horizon planning, tool management, and natural interaction. Emerging benchmarks such as HeroBench and FieldWorkArena represent early steps in this direction; however, they still face methodological challenges, including the simulation-to-real gap and vulnerability to benchmark gaming [80, 88]. In parallel, the development of initial protocols for

Table 4 Research roadmap for Agentic AI

Timeframe	Research Axis	Key Objective & Actions	Target Application Areas
Short-term (1–2 years)	■ Persistent Memory & Infrastructure	■ Development of hierarchical memory frameworks ■ Design of interactive and process-oriented benchmarks (HeroBench, FieldWorkArena) ■ Creation of initial protocols for coordination among agents	■ Laboratory and controlled-environment evaluation
Short-term (1–2 years)	■ Foundational Security	■ Design of defense mechanisms against prompt injection ■ Implementation of sandboxing ■ Agent authentication	■ All LLM-based domains
Medium-term (3–5 years)	■ Reasoning & Learning	■ Integration of multi-step reasoning and self-reflection with Small Language Models (SLMs) ■ Integration of online learning with long-term memory	■ Robotics ■ Medical systems ■ Education
Medium-term (3–5 years)	■ Advanced Collaboration & Security	■ Development of safe, scalable multi-agent interaction architectures ■ Design of formal threat models	■ Medical system ■ Finance ■ Legal
Medium-term (3–5 years)	■ Transparency & Auditing	■ Design of transparency frameworks for decision-making ■ Basic auditing and accountability mechanisms	■ Public Services ■ Healthcare ■ Education
Long-term (more than 5 years)	■ Autonomous Agency	■ Deployment of fully autonomous systems for scientific discovery and materials/drug design ■ Agentic AI as an active research partner in exploration	■ Natural Sciences ■ Pharmaceuticals ■ Materials engineering

dynamic coordination and experience sharing among agents will begin in this period to prevent unpredictable behaviors in multi-agent settings [21]. At the security layer, research should focus on establishing foundational defensive mechanisms, including sandboxing, agent authentication, and protection against prompt injection attacks.

8.2 Mid-Term (3–5 Years): Integration and Systemic Maturity

During this period, the integration of multi-step reasoning approaches and self-reflection mechanisms with smaller and more efficient language models is expected to reach maturity. This direction can both improve reasoning accuracy and reduce energy consumption and computational resource requirements an objective that has gained increasing importance with the emergence of Small Language Models (SLMs) as a key component of the future Agentic AI ecosystem [91]. At this stage, research should move toward developing architectures that support safe and scalable interactions among agents, both in virtual environments and in real-world domains such as robotics or medical systems [23]. Additional priorities include the integration of online learning with long-term memory and the design of formal threat models for multi-agent ecosystems. From an application perspective, controlled pilot deployments are anticipated in fields such as healthcare for example, as multi-purpose decision-support assistants and education, where agents may function as simulators and personalized companions for experiential learning [110, 113]. Achieving success in this phase will require the development of frameworks that ensure decision-making transparency along with initial mechanisms for auditing and accountability [77, 78].

8.3 Long-Term (More than 5 Years): Autonomous Agency, Sustainability, and Value Alignment

In the long term, the research vision moves toward the deployment of fully autonomous systems in sensitive and complex domains such as scientific discovery and the design of new materials or pharmaceuticals. In this context, Agentic AI functions as an active partner in the discovery process, assisting researchers in generating hypotheses, designing experiments, and analyzing complex data [103]. At this stage, the fundamental ethical and societal challenges associated with autonomous systems must be comprehensively addressed. Research should lead to the development of robust frameworks for value alignment with human norms, continuous auditing, and transparent accountability, ensuring that these systems operate in ways that maintain and strengthen public trust. Additionally, the environmental sustainability of large-scale agent ecosystems will become a central concern. Future work must therefore explore methods to reduce the ecological footprint of AI infrastructures while also establishing large-scale ethical governance mechanisms capable of managing increasingly autonomous and interconnected agent systems.

9 Conclusion

Agentic AI has taken a step beyond general purpose language models and classical agents by combining multi step reasoning, persistent memory, multi agent interaction, and dynamic tool use to provide a novel foundation for developing autonomous intelligent systems. Agentic architectures require careful design not only at the component level,

spanning perception and role allocation to planning, action, and online learning, but also in the supporting infrastructure, including state and memory management, scalable orchestration, and safeguarding frameworks. Progress in this domain is not possible without precise and reproducible evaluation criteria. The development of interactive benchmarks, attention to the internal processes of agents, and the design of comparable protocols are essential future requirements. Alongside this, applications of Agentic AI across diverse domains such as scientific discovery, education, finance, and robotics demonstrate the transformative potential of this technology for industries and interdisciplinary research. Nevertheless, challenges such as emergent behaviors, scalability, debugging of multi agent systems, data security and governance, social responsibility, and energy costs remain. To address these obstacles, more integrated frameworks such as Agentic RAG, agent chains, and formal threat models are increasingly needed.

In sum, Agentic AI stands at the threshold of becoming a dominant paradigm in artificial intelligence. This field not only offers a new pathway for enhancing the capabilities of language models but also provides a foundation for fundamental research in multi agent interaction, self-improving learning, and the safeguarding of intelligent systems. It is expected that through advancing future studies in transparent architecture design, reproducible evaluation protocol development, and the integration of ethical and environmental considerations, Agentic AI will be deployed sustainably and reliably across scientific, industrial, and societal domains.

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Declarations

Ethical approval This article does not contain any studies with human participants or animals performed by any of the authors.

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